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Review article

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The therapeutic potential of orphan adhesion G-protein-coupled receptors

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Abstract

Adhesion G-protein-coupled receptors (aGPCRs) are essential membrane proteins that contribute to diverse human pathologies, including cancer, psychiatric disorders and autoimmune diseases. A defining feature of these receptors is the large N terminus including the GPCR autoproteolysis-inducing domain, which mediates autoproteolysis and enables complex modes of activation through both *cis* and *trans* signalling. Recent breakthroughs in structural biology, in particular, cryogenic electron microscopy structures of aGPCRs, have transformed our understanding of receptor activation and facilitated development of small-molecule agonists and antibody-based strategies, which hold considerable promise for therapeutic intervention. In this Review, we summarize the roles of aGPCRs in human disease and in phenotypes characterized in animal models. We consolidate current insights into aGPCR structure–function relationships, highlight how distinct activation mechanisms are already being harnessed to modulate receptor activity in vitro and in vivo, and discuss how these principles could be leveraged for future therapeutic targeting.

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Introduction

Adhesion G-protein-coupled receptors (aGPCRs) constitute a diverse class of membrane proteins that have been implicated in multiple human pathologies including cancer, psychiatric disorders, autoimmune diseases and metabolic dysregulation, rendering them highly interesting drug targets. Over the years, aGPCRs have been ascribed various names (Table 1). In 2015, a new classification assigned mammalian aGPCRs into nine distinct families, with gene symbols beginning with ADGR[1].

aGPCRs exhibit many structural and functional features that are characteristic of GPCRs in general, including a seven-transmembrane domain (7TMD), an extracellular N terminus, and an intracellular C terminus (Fig. 1). Ligand binding within the 7TMD stabilizes an active conformation, enabling interaction with G proteins and subsequent intracellular signal transduction. Termination of signalling can occur via arrestin recruitment, as observed for many GPCRs. In addition to these canonical features, aGPCRs incorporate a range of atypical properties that are also found in other classes of membrane receptors. These include (1) adhesion to extracellular and intracellular ligands, suggesting bidirectional (*cis* versus *trans*) signalling mechanisms akin to those of integrins; (2) responsiveness to a broad spectrum of mechanical stimuli, a property shared with certain ion channels, integrins and other GPCRs; (3) interaction with other membrane and adapter proteins beyond G proteins and arrestins, as reported for Frizzled receptors; and (4) activation through an intramolecular agonist sequence[2,3]. These noncanonical features are enabled by distinctive structural elements such as the GPCR autoproteolysis-inducing domain (GAIN domain)[4] and further functional domains that extend the N-terminal and C-terminal regions (Fig. 1a). The GAIN domain mediates cleavage at a highly conserved GPCR proteolytic site (GPS), located within the extracellular N terminus in close proximity to the first transmembrane domain. Cleavage at the GPS divides the receptor into an N-terminal fragment (NTF) and a C-terminal fragment (CTF). The CTF comprises the intracellular C terminus, the 7TMD and the residual N-terminal amino acids. For several aGPCRs, the CTF displays constitutive activity[5,6], a property that enabled the discovery of the intramolecular agonistic sequence[2,3]. The agonistic sequence within aGPCRs was originally termed, and is still frequently referred to as, a tethered agonist[2,3], in analogy to the self-activating peptide of protease-activated receptors (PARs). However, the term tethered agonist more generally denotes ligands that are covalently attached but not necessarily intrinsic to the receptor polypeptide, as exemplified by the rhodopsin ligand 11-*cis*-retinal. To avoid this ambiguity, we favour the broader designation 'intramolecular ligand' for aGPCRs. Upon its discovery, we proposed the specific name 'Stachel' for this sequence[2], which obviates the need for such semantic distinctions. Accordingly, Stachel will be used throughout this Review for clarity.

Full understanding of the complex processes attributed to aGPCR function is essential for rational drug design with minimal side effects. Mechanisms of particular interest involve the extracellular N-terminal region, which serves a dual role: as an acceptor of allosteric modulators that trigger intracellular cis signalling via the 7TMD[7], and as a ligand for other proteins, enabling *trans* signalling[8]. A range of soluble and insoluble ligands, as well as mechanical forces, can control N terminus-mediated *cis* activation, often involving the Stachel sequence.

Synthetic peptides based on the Stachel sequence have been used as agonists in cell culture and animal models, but limited specificity and potency have hindered therapeutic development. Broader ligand screens identified synthetic small molecules, but early candidates generally lacked the specificity needed for clinical efficacy[9]. A breakthrough came with cryogenic electron microscopy (cryo-EM) resolution of active 7TMD structures[10], validating prior functional data and enabling in silico drug screening.

Recent advances in aGPCR drug discovery establish aGPCRs as promising targets for translational research. This Review outlines their contribution to human pathologies and physiological implications from animal models. It provides an overview of currently available endogenous and synthetic modulators and discusses how the structural and functional features of these receptors can be harnessed for therapeutic development.

aGPCRs in health and disease

aGPCRs are attractive drug targets owing to their essential roles in physiological regulation. Mutations, genetic variants or altered expression are associated with a wide range of human diseases. Much of our current understanding derives from model organisms such as *Caenorhabditis elegans*, *Drosophila melanogaster*, *Danio rerio* and *Mus musculus*. These systems permit rapid in vivo manipulation of aGPCRs and reveal the physiological consequences of receptor perturbation. Complementary cell culture studies allow dissection of the underlying molecular mechanisms.

Despite differences in complexity, some functions are conserved across species when orthologues exist. *C. elegans* and *Dr. melanogaster* encode only three and six aGPCR-like genes, respectively[11,12]. By contrast, *D. rerio* encodes orthologues for nearly all 33 human aGPCR genes. Following a teleost-specific whole-genome duplication, several paralogues exist, bringing the total to 59 in some annotations[13]. *M. musculus* and humans encode 30 and 33 orthologues, respectively[1]. Each species also expresses multiple splice variants, which expand receptor diversity beyond the number of protein-coding genes[14,15].

To date, seven human diseases have been conclusively attributed to specific mutations in the respective aGPCR genes, namely bilateral frontoparietal polymicrogyria (BFPP)[16], microcephaly and rhombencephalosynapsis[17], craniorachischisis[18], lethal arthrogryposis multiplex congenita[19], Usher syndrome type II (USH2)[20], spastic paraparesis[21], and vibratory urticaria[22] (Table 1). Several other associations have been established, linking 17 out of 33 human aGPCRs to human pathologies. Multiple phenotypes observed in animal models resemble those seen in human disease, underscoring their translational relevance. Nevertheless, not all phenotypes identified in humans have been examined in animal models, nor have all phenotypes found in aGPCR-deficient animals been observed in humans. This Review (and specifically Table 1) aims to highlight the current gaps and hopes to inspire new investigations to address them. The assembled phenotypic evidence should further help researchers select the most suitable animal model for the pathology or receptor under study.

Physiological and pathophysiological insights from animal models

Extensive work across *C. elegans*, *Dr. melanogaster*, *D. rerio* and *M. musculus* underscores essential and conserved roles for aGPCRs in development, tissue homeostasis and inter-organ communication. Loss-of-function phenotypes illuminate mechanisms relevant to human disease associations, whereas transgenic and conditional models reveal organ-specific and species-specific roles that often align across systems. Limitations in the conservation of a given phenotype arise from the repertoire of aGPCRs encoded in each species and the presence

Table 1 | Adhesion GPCR human disease associations and phenotypes in KO animal models

Class	ADGR	Alternative name	Human disease (association)	Mouse KO	Zebrafish KO	Fruitfly KO	Nematode KO	Refs.
A	1	GPR123	Unknown	Unknown	Unknown	n.p.	n.p.	–
	2	GPR124, TEM5	Unknown	Unknown	Brain vascularization, BBB development	CG15744,remoulade: axonal morphogenesis CG11318/mayo: midgut development	n.p.	42,47,48
	3	GPR125, TEM5L, PGR21	Unknown	Bone strength, infertility	Disrupted embryonic development		n.p.	43,50,92, 93,102
B	1	BAI1, GDAIF	Unknown	Behaviour deficits, epilepsy, suppression of medulloblastoma formation	Unknown	n.p.	n.p.	55,108
	2	BAI2	Progressive spastic paraparesis	Unknown	Unknown	n.p.	n.p.	21
	3	BAI3	Psychiatric disorders	Unknown	Unknown	n.p.	n.p.	125,126
C	1	CELSR1,	Neural tube defects, psychiatric disorders, epilepsy	Neural tube defects, compromised skin integrity and function	Unknown	Fmi, Stan: neural development, tissue organization	FMI-1: neural development, axon guidance, body size regulation, control of extracellular matrix	18,32–38, 59,77,98, 113,114, 122–124
	2	CELSR2, EGFL2,	Psychiatric disorders	Impaired pancreatic β-cell function	Unknown			
	3	CELSR3	Unknown	Impaired pancreatic β-cell function	Unknown			
D	1	GPR133, PGR25	Bone strength, body height, glioblastoma	Reduced muscle strength, bone strength, infertility, glioblastoma growth	Unknown	n.p.	n.p.	89–91, 101,109, 144–146
	2	GPR144, PGR24	Unknown	Unknown	Unknown	n.p.	n.p.	–
E	1	EMR1, F4/80, TM7LN3	Unknown	Regulatory T cell dysfunction	Unknown	Mthl1: altered innate immune responses	n.p.	12,64
	2	EMR2, CD312	Vibratory urticaria, rheumatoid arthritis	Unknown	Unknown	n.p.	n.p.	22, 131,132
	3	EMR3	Unknown	Unknown	Unknown	n.p.	n.p.	–
	4	EMR4	Unknown	Unknown	Unknown	n.p.	n.p.	–
	5	CD97, TM7LN1	Rheumatoid arthritis, multiples sclerosis	Improved antibacterial host defence, thyroid cancer progression	Unknown	n.p.	n.p.	67,104, 130–132
F	1	GPR110, PGR19	NAFLD	NAFLD	Unknown	n.p.	n.p.	80
	2	GPR111, PGR20, HGPCR35	Unknown	Unknown	Unknown	n.p.	n.p.	–
	3	GPR113, PGR23	Unknown	Unknown	Unknown	n.p.	n.p.	–
	4	GPR115, PGR18	Unknown	Enamel mineralization	Unknown	n.p.	n.p.	97
	5	GPR116, Ig-Hepta	Obesity, glucose tolerance	Glucose tolerance, impaired pancreatic islet development, pulmonary surfactant secretion, regulation of acid–base homeostasis in the kidney, vessel formation	Unknown	n.p.	n.p.	69,74–76, 84–87,138
G	1	GPR56, TM7LN4, TM7XN1, BFPP, BPPR	Brain malformation BFPP, depression, epilepsy, Alzheimer's disease, multiple sclerosis, myocardial infarction, rheumatoid arthritis	Brain development defects, cortical malformation, central hypomyelination, traumatic brain injury pathology, cardiac dysfunction, defective platelet function, ferroptosis-induced liver injury, muscle adaptation to mechanical overload	Central hypomyelination, increased peripheral myelin thickness	n.p.	n.p.	16,45,46, 51–53,63, 70,71,81, 88,110–112, 117,118, 127–129, 134,137

Table 1 (continued) | Adhesion GPCR human disease associations and phenotypes in KO animal models

Class	ADGR	Alternative name	Human disease (association)	Mouse KO	Zebrafish KO	Fruitfly KO	Nematode KO	Refs.
G (continued)	2	GPR64, HE6, EDDM6, TM7LN2	Infertility	Infertility	Unknown	n.p.	n.p.	99,100,147
	3	GPR97, PB99, PGR26	Pneumonia	B-Lymphocytes number, suppressing psoriasis, renal interstitial fibrosis, immune cell infiltration in acute kidney injury	Unknown	n.p.	n.p.	65,66,80, 82,83,135
	4	GPR112, PGR17, Gm367	Unknown	Unknown	Unknown	n.p.	n.p.	–
	5	GPR114, PGR27, Gm1109	Unknown	Unknown	Unknown	n.p.	n.p.	–
	6	GPR126, DREG, VIGR, Gm222, PR126, APG1, PS1TP2	Arthrogryposis multiplex congenita, myelination defects, scoliosis, body length	Joint contractures linked to human arthrogryposis multiplex congenita, Schwann cell myelination and peripheral nerve repair defect, scoliosis and pectus excavatum, body length, bone strength, cardiac development	Schwann cell myelination defect, altered inner-ear morphogenesis, cardiac trabeculation defect	n.p.	n.p.	19,24,44, 61,62,72, 94–96, 140–143
	7	GPR128	Unknown	Unknown	Unknown	n.p.	n.p.	–
L	1	LPHN1, CL1, CIRL1, LEC2	Obesity	Obesity, type 2 diabetes	Unknown	Mechanosensation in sensory neurons, planar cell polarity, neuroblast number	Disrupted embryonic development, infertility	26–30, 39–41, 78,79
	2	LPHN2, CL2, CIRL2, LEC1	Brain malformation	Altered synaptic transmission, inner-ear hair cell defects, improved myocardial repair	Unknown	n.p.	Disrupted embryonic development	17,31,60,73
	3	LPHN3, CL3, CIRL3, LEC3	Unknown	Altered synaptic transmission, ADHD	Unknown	n.p.	n.p.	54
	4	ELTD1, ETL	Heart failure	Cardiac hypertrophy, vessel formation	Unknown	n.p.	n.p.	68,69
V	1	VLGR1, FEB4, GPR98, MASS1, USH2B, USH2C	Usher syndrome type II, hearing loss, epilepsy	Epilepsy, defective inner-ear hair development, defective photoreceptors	Unknown	n.p.	n.p.	20,56–58, 119,120

Receptors are sorted according to class. BBB, blood–brain barrier; BFPP, bilateral frontoparietal polymicrogyria; GPCR, G-protein-coupled receptor; KO, knockout; NAFLD, non-alcoholic fatty liver disease; n.p., not present.

of the corresponding organ systems. Accordingly, *C. elegans* and *Dr. melanogaster* cannot be expected to display all vertebrate phenotypes observed in mouse and human. Nevertheless, fundamental roles of aGPCRs in cell biology, and particularly in the nervous system, can be detected in these organisms and, owing to their simplicity, can be screened rapidly. This could be exploited for pharmacological screening; however, in *C. elegans*, such approaches have so far been applied predominantly to anti-infective compounds[23], whereas in vivo compound screens targeting aGPCRs have largely been restricted to *D. rerio*[24,25]. This vertebrate model has proven highly valuable in this regard. Extending the currently established phenotypic repertoire would undoubtedly open new opportunities for in vivo compound identification and validation, especially given that *D. rerio* encodes nearly all aGPCRs found in humans[13]. By contrast, unbiased compound screens in mice are neither efficient nor compatible with most international standards of animal welfare. Nonetheless, the genetic and physiological proximity of *M. musculus* to humans makes it an essential model for investigating specific cell-level and tissue-level functions of aGPCRs linked to patient cohorts. Disease-specific or receptor-deficient mouse models can moreover be used to evaluate candidate aGPCR-targeting compounds expected to ameliorate disease burden.

The following paragraphs provide a brief overview of aGPCR-associated phenotypes observed across animal models, with the aim

of guiding drug development and screening efforts using these in vivo systems.

C. elegans. *C. elegans* encodes three aGPCR orthologues: LAT-1, LAT-2 (latrophilin) and FMI-1 (*Flamingo*/CELSR)[26]. LAT-1 regulates early cell polarity and embryonic axis formation via the intramolecular agonist and cAMP signalling[27,28] and, via its N terminus, controls germline development and sensory morphogenesis affecting copulation behaviour and neuronal physiology[29,30]. By contrast, LAT-2 deletion shows no overt phenotype[31]. FMI-1 shapes neural circuitry, including dendrite self-avoidance[32], axon guidance[33] and synaptogenesis[34], with additional roles in body size and extracellular matrix composition[35].

Dr. melanogaster. *Dr. melanogaster* studies highlight aGPCR roles in morphogenesis and systemic physiology. *Flamingo (Fmi)/Starry night (Stan)* contributes to neural development[36], tissue organization[37] and cell competition[38]. *Cirl* modulates mechanosensation (touch, sound and nociception)[39,40] and governs planar cell polarity and neuroblast pool size via interaction with the Toll-like receptor Tollo[41]. *Remoulade* regulates axonal morphogenesis[42]. *Mayo* is essential for midgut development and enterocyte proliferation; its loss disrupts potassium homeostasis, causing hyperkalaemia and tachycardia, linking gut–heart physiology[43]. *mthl1* contributes to innate antitumour responses[12].

D. rerio. Despite numerous orthologues, functional assignments cluster to a few receptors. Adgrg6/Gpr126 is essential for Schwann cell myelination and inner-ear morphogenesis via cAMP signalling modulated by extracellular matrix cues (summarized in ref. 24). It also regulates cardiac trabeculation and endocardial–myocardial communication, with an N-terminal region mediating this activity[44]. Adgrg1/Gpr56 controls central and peripheral myelination: oligodendrocyte loss causes hypomyelination[45], whereas Schwann cell loss increases myelin thickness via SREBP1-linked lipid biosynthesis[46]. $G_{12/13}$–RhoA signalling is engaged in both contexts. Adgra2/Gpr124 is required for brain angiogenesis[47] and blood–brain barrier formation[48] as a RECK-dependent co-receptor for WNT7A/WNT7B, with intracellular motifs recruiting dishevelled (DVL), DLG4 and MAGI3 to potentiate Wnt/β-catenin signalling[49]. Adgra3/Gpr125 drives convergence and extension during gastrulation by recruiting DVL to the membrane, aligning with Wnt/planar cell polarity phenotypes[50].

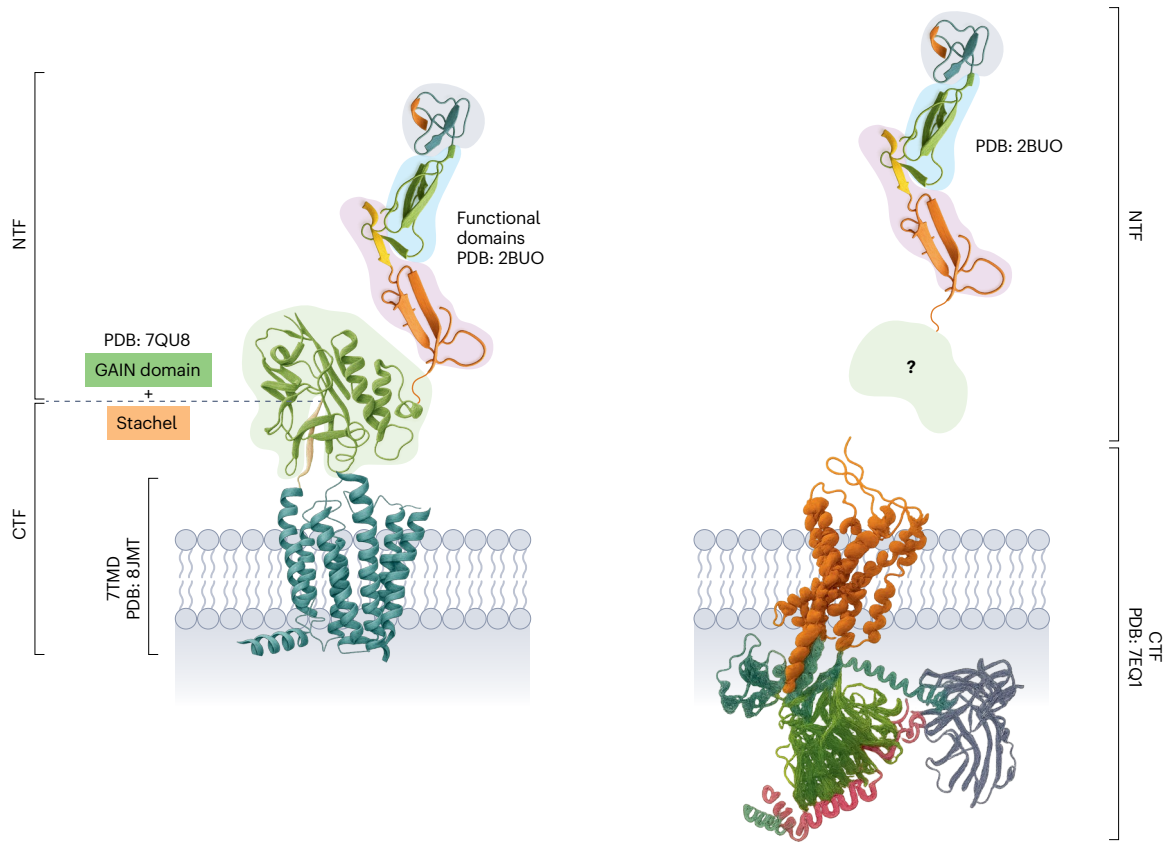


Fig. 1 | Structural features of adhesion GPCR. A hallmark feature of adhesion G-protein-coupled receptors (GPCRs) is their extraordinarily large extracellular region, which consists of several functional domains. The GPCR autoproteolysis-inducing (GAIN) domain cleaves the receptor into an N-terminal fragment (NTF) and a C-terminal fragment (CTF), which remain non-covalently attached. Additional functional domains can be found upstream of the GAIN domain. The N-terminal tip of the CTF marks the beginning of the intramolecular agonist sequence, known as the 'Stachel'. In the full-length receptor, the Stachel remains buried within the GAIN domain. Upon dissociation of the NTF, the Stachel changes its conformation as it engages with the seven-transmembrane domain (7TMD) bundle. As no full-length structure is available for an adhesion GPCR, the structural assembly is compiled based on single structures of ADGRL3/LPHN3 7TMD in the apo state (PDB: 8JMT), the GAIN domain of ADGRG3/GPR97 (PDB: 7QU8), and the EGF domains of ADGRE5/CD97 (PDB: 2BUO). The CTF structure shown is from ADGRG5/GPR114 (PDB: 7EQ1). Of note, there is no resolved structure of a GAIN domain without the Stachel sequence available (indicated with a question mark in the figure).

***M. musculus*.** Knockout (KO) and conditional models exist for all 31 mouse aGPCR orthologues, demonstrating roles in neurodevelopment and sensation, immunity, cardiovascular biology, metabolism, musculoskeletal physiology and cancer.

In the nervous system, *Adgrg1*/*Gpr56* preserves the pial basement membrane and neuron–glia interactions to prevent cobblestone-like cortical malformations[51,52]. Receptor loss further aggravates traumatic brain injury with larger lesions, worse functional recovery, and blood–brain barrier disruption[53]. *Adgrl3*/*Lphn3* shapes excitatory and inhibitory synapses, and its deletion yields hyperactivity, impulsivity and memory deficits reminiscent of ADHD-like phenotypes[54]. *Adgrb1*/*Bai1* deficiency reduces brain weight, increases apoptosis, alters social behaviours, and heightens seizure susceptibility, consistent with dosage-sensitive neurodevelopmental roles[55]. Deficiency in *Adgrv1*/*Vlgr1* causes audiogenic seizures and defects in inner-ear ankle links and photoreceptor links[56–58], whereas *Adgrc1*/*Celsr1* mutants show neural tube closure failure and disordered hair-cell orientation, reflecting planar cell polarity disruption[59]. Hair-cell-specific *Adgrl2*/*Lphn2* loss impairs mechanotransduction currents and balance, both rescued by receptor re-expression, indicating a force-sensitive role in vestibular physiology[60]. In addition, *Adgrg6*/*Gpr126* is essential for Schwann-cell myelination[61] and repair[62]. Mutants display joint contractures that parallel human arthrogryposis multiplex congenita[19]. *Adgrg1*/*Gpr56* also promotes central myelination by supporting oligodendrocyte precursor proliferation[63].

In immunity, *Adgre1*/*Emr1* is required to induce efferent regulatory T cells and robust delayed-type hypersensitivity responses[64]. ADGRG3/GPR97 modulates B cell populations[65] and constrains psoriasiform autoimmunity in KO studies[66], whereas *Adgre5*/*Cd97* deletion improves antibacterial host defence[67], underscoring receptor-specific and context-specific immune roles for aGPCRs.

Cardiovascular studies indicate that *Adgrl4*/*Eltd1* loss exacerbates pressure-overload hypertrophy[68] and, together with *Adgrf5*/*Gpr116* deletion, produces vascular malformations[69]. Cardiomyocyte-specific loss of *Adgrg1*/*Gpr56* accelerates dysfunction under chronic pressure overload, blunts compensatory hypertrophy, heightens inflammation and increases mortality[70]. Platelet ADGRG1/GPR56 mediates collagen responses; global KO extends bleeding times and destabilizes clot formation under shear stress[71]. ADGRG6/GPR126 is required for cardiac development[72], with defects linked to mitochondrial dysfunction and impaired endocardial–myocardial signalling[44]. Meanwhile, ADGRL2/LPHN2 marks and enables enrichment of cardiac progenitor cells, suggesting a role in early cardiac lineage specification[73].

In metabolism and organ physiology, *Adgrf5*/*Gpr116* deletion alters glucose homeostasis and insulin sensitivity and disrupts islet development[74–76]. Disruption of *Adgrc2*/*Celsr2* and *Adgrc3*/*Celsr3* perturbs planar cell polarity in endocrine cells, impairing β-cell function[77]. Whole-body or hypothalamus-specific *Adgrl1*/*Lphn1* loss increases food intake and reduces activity, driving obesity and type 2 diabetes-like features[78,79]. Hepatic *Adgrf1*/*Gpr110* is downregulated by high-fat diet, whereas overexpression promotes steatosis[80]. Hepatocyte-specific *Adgrg1*/*Gpr56* loss exacerbates ferroptosis-induced liver injury via impaired CD36-mediated lipid handling[81]. This phenotype is rescued by the endogenous agonist 17α-hydroxy pregnenolone, which suppresses ferroptosis. In the kidney, *Adgrg3*/*Gpr97* upregulation during hypertensive injury promotes interstitial fibrosis through RhoA and TGF-β signalling[82], whereas its loss mitigates fibrosis and inflammation and improves outcomes in cisplatin-induced injury[83]. In addition, ADGRF5/GPR116 in kidney A-intercalated cells regulates V-ATPase trafficking and acid–base homeostasis[84]. In the lung, ADGRF5/GPR116 in alveolar type II cells is essential for surfactant turnover, and KO causes surfactant accumulation, structural abnormalities and progressive respiratory failure[85–87].

Within the musculoskeletal system, *Adgrg1*/*Gpr56* is dispensable for baseline muscle formation but required for overload-induced hypertrophy[88]. ADGRD1/GPR133 has a role in the musculoskeletal system wherein its KO significantly attenuates the muscle-strengthening effects of 5α-dihydrotestosterone[89] and decreases mechanical-load-induced bone strength[90]. ADGRD1/GPR133 synergistically governs skeletal muscle, osteoblast and osteoclast activity through the Gs/cAMP pathway[89–91], whereas ADGRA3/GPR125 promotes osteoclastogenesis in vitro[92], yet global deletion reduces bone mineral density in vivo[93], indicating context-dependent effects. Osteoblast-specific *Adgrg6*/*Gpr126* loss shortens body length and reduces bone mineralization via cAMP–CREB signals, with collagen IV as a ligand[94]. Cartilage and connective-tissue deletion induces scoliosis, pectus excavatum[95] and endplate-oriented disc herniations accompanied by STAT3 activation and increased disc stiffness[96]. Additionally, *Adgrf4*/*GPR115* regulates enamel mineralization through carbonic anhydrase 6 in ameloblasts[97], and ADGRC1/CELSR1 orients planar cell divisions in epidermis to maintain barrier integrity[98].

Reproductive phenotypes include male infertility owing to obstructive azoospermia in *Adgrg2*/*Gpr64* KOs, reflecting defective fluid handling in efferent ducts[99,100]. Female *Adgrd1*/*Gpr133* KOs are sterile owing to failure to overcome the ampullary–isthmic junction restraint, causing embryo retention[101]. *Adgra3*/*Gpr125* KOs develop male obstructive azoospermia[102] and, in approximately 44% of females, imperforate vagina unresponsive to 17β-oestradiol, leading to infertility[93].

In cancer models, ADGRE5/CD97 promotes tumour growth, migration and angiogenesis in colorectal, gastric and thyroid settings[103,104]; ADGRF5/GPR116 supports breast cancer metastasis and extracellular-matrix-associated migration[105,106]; *Adgrf1*/*Gpr110* acts as a proto-oncogene in leukaemia[107]; ADGRB1/BAI1 functions as a tumour suppressor by stabilizing p53 and inhibiting angiogenesis[108]; and *Adgrd1*/*Gpr133* is upregulated in glioblastoma and required for tumour cell survival and hypoxic adaptation in vitro and in xenografts[109]. Together, these data position aGPCRs as modulators of development and disease across systems. In particular, the mouse phenotypes aligning to human genetics and pathology could highlight promising indications for therapeutic intervention.

Human monogenetic diseases caused by mutations in aGPCRs

Among human pathologies directly caused by protein-coding mutations in aGPCRs, disorders of the nervous system are most prominent. Mutations affecting nervous system development are often embryonic or postnatal lethal. Despite advances in understanding underlying mechanisms, treatment remains limited to genetic counselling.

Nonsense and missense mutations in *ADGRG1*/*GPR56*, *ADGRL2*/*LPHN2*, *ADGRC1*/*CELSR1* and *ADGRG6*/*GPR126* cause lethal developmental disorders. For example, ADGRG1/GPR56 loss-of-function mutations cause BFPP, a severe congenital brain malformation characterized by thickened cortex and irregular gyri[16]. Patients present with developmental delay, intellectual disability, seizures and spasms. All 14 known BFPP mutations yield similar severity[110], although mechanisms differ from impaired trafficking[111] to disruption of N-terminal collagen III binding[112]. In addition, a heterozygous missense mutation in *ADGRL2*/*LPHN2* was reported in a fetus with extreme microcephaly

and rhombencephalosynapsis[17]. The resulting $Leu_{1262}His$ substitution is located in the C terminus of the receptor. Although receptor expression was unaffected, intracellular calcium release upon α-latrotoxin stimulation was reduced, impairing adhesion and migration. Furthermore, six non-synonymous *ADGRC1/CELSR1* mutations have been identified in fetuses with craniorachischisis, a severe neural tube defect[18]. Four mutations reduced cell surface expression, but no other functional analyses were performed. Additional missense variants in neural tube defect-affected pregnancies remain uncharacterized[113]. In spina bifida, TG dinucleotide repeats truncate the N terminus, preventing membrane localization and impairing planar cell polarity via defective VANGL2 recruitment[114]. Finally, *ADGRG6/GPR126* nonsense and missense mutations have been found in fetuses and a neonate with lethal arthrogryposis multiplex congenita[19], marked by craniofacial abnormalities, pulmonary hypoplasia and pterygia. Myelin staining revealed absent peripheral myelin. The missense mutation within the GAIN domain produced a non-cleavable receptor, implicating disrupted autoproteolysis.

Mutations in ADGRV1/VLGR1, ADGRB2/BAI2 and ADGRE2/EMR2 are compatible with life but induce disability. For example, four *ADGRV1/VLGR1* mutations cause USH2, an autosomal recessive disorder with congenital hearing loss and progressive retinitis pigmentosa[20]. Mutations cluster in the large N terminus or C terminus, probably disrupting complex assembly[115]. In addition, a de novo *ADGRB2/BAI2* mutation (R1465W) was reported in a patient with spastic paraparesis[21]. Although full-length expression was preserved, CTF-only constructs doubled in abundance, with increased NFAT luciferase activity, suggesting gain-of-function. Finally, the ADGRE2/EMR2 mutation C492Y causes vibratory urticaria, in which mechanical stimulation triggers erythematous, pruritic hives[22]. Located in the GAIN domain, the mutation promotes NTF shedding and enhances mast cell degranulation. Both wild-type (WT) and mutant receptors activate phospholipase C-dependent calcium signalling, but mutant-induced responses are markedly stronger[116].

aGPCRs and human disease contributions

Beyond monogenic disorders, many intronic variants in aGPCR genes have been associated with complex human diseases, although most require functional validation in animal models. Dysregulated expression patterns represent another level of disease involvement. Numerous studies report altered receptor expression in pathological states, but whether these changes are causal or compensatory often remains unclear. In contrast to lethal developmental defects caused by monogenic mutations, aGPCR contributions to complex diseases in living populations present opportunities for therapeutic intervention. Receptors can be targeted directly or via their downstream signalling cascades, as is already established for other GPCR-mediated pathologies.

The following sections provide an overview of notable human diseases associated with altered expression or function of specific aGPCRs.

Neurological disorders. aGPCRs have been implicated in a range of neurological and psychiatric conditions through a combination of human genetics, transcriptomics and animal models. Several receptors show the strongest human links, although for many the evidence remains associative and requires functional follow-up in model systems.

ADGRG1/GPR56 has surfaced in diverse clinical contexts. A recent study has reported significantly decreased expression in patients with depression who died by suicide, and expression was upregulated in patients responding to duloxetine treatment, suggesting a possible role in affective disorders and antidepressant response[117]. Links between aGPCR variation and epilepsy are suggestive but not definitive. Several studies have reported potential involvement of *ADGRG1/GPR56* in epilepsy[118], yet most reported phenotypes occur alongside structural brain malformations rather than as isolated epilepsy. *ADGRV1/VLGR1* has been implicated in myoclonic epilepsy[119] and febrile seizures[120] in some reports, but these human cases typically involve larger, complex deletions at chromosome 5q14.3 (which include additional genes) and are often partial deletions of *ADGRV1/VLGR1* rather than isolated gene lesions. A separate family study has mapped febrile-seizure linkage to 5q14.3 but found no clear disease-causing mutation in *ADGRV1/VLGR1*, challenging a simple causal role[121]. Further supportive evidence for the involvement of *ADGRV1/VLGR1* in seizures stems from a ClinGen Clinical-Validity Framework study reporting *ADGRC1/CELSR1* variation in 6 of 356 patients with partial epilepsy[122].

Several aGPCRs are recurrently dysregulated in psychiatric disorders. *ADGRC1/CELSR1* and *ADGRC2/CELSR2* rank among the most consistently altered genes in datasets for autism spectrum disorder, schizophrenia and bipolar disorder[123]. De novo rare variants have been reported in multiple aGPCR genes in autism cohorts, including *ADGRB1-3/BAI1-3*, *ADGRV1/VLGR1*, *ADGRC1,3/CELSR1,3*, *ADGRL1-3/LPHN1-3* and *ADGRL4/ELTD1* (ref. 124). Genome-wide association studies have linked intronic single-nucleotide polymorphisms (SNPs) in *ADGRB3/BAI3* to autism, schizophrenia and Alzheimer's disease[125,126], and intronic or exonic SNPs in *ADGRG1/GPR56* have been associated with Alzheimer's disease[127] and multiple sclerosis[128], respectively.

Expression studies provide additional, disease-relevant evidence. *ADGRG1/GPR56* was reported as downregulated in microglia isolated from white and grey matter of deceased patients with multiple sclerosis compared with non-pathological brains[129]. A focused post-mortem investigation of ADGRE5/CD97 and its ligand CD55 found ADGRE5/CD97 absent from morphologically unaffected white matter but increasingly expressed in infiltrating immune cells in pre-active and active multiple-sclerosis lesions; CD55 was present on endothelial cells at all stages and additionally in infiltrating immune cells[130]. The authors concluded that ADGRE5–CD55 interactions may facilitate activation and migration of infiltrating immune cells in multiple-sclerosis lesions.

Immune and inflammatory diseases. Several aGPCRs are expressed in immune cells and have been implicated in immune defence and inflammatory processes. Although the functional contribution of selected aGPCRs to immune cell function has been intensively studied in vitro and in animal models, so far, there has been only one direct variant-disease link described for vibratory urticaria, mediated by a missense mutation in *ADGRE2/EMR2* (ref. 22). However, other inflammatory disease states are accompanied by changes in specific aGPCR expression levels. For instance, *ADGRE2/EMR2* and *ADGRE5/CD97* are upregulated in synovial tissue from patients with rheumatoid arthritis[131,132]. Interestingly, blocking ADGRE5/CD97 with an antibody ameliorates joint inflammation in experimental arthritis mouse models[133], indicating that upregulation of the receptor contributes to disease progression rather than being a sign of physiological compensation. Rheumatoid arthritis patients also present with elevated levels of soluble N-terminal parts of aGPCRs ADGRG1/GPR56 (ref. 134) and ADGRE5/CD97 (ref. 131). Moreover, the expression of *ADGRG3/GPR97* is significantly increased in blood cells from patients with pneumonia and can be induced by treating blood cells from healthy donors with lipopolysaccharide[135]. As activation of ADGRG3/GPR97

enhances antimicrobial functions, it can be speculated that induced expression of the receptor is a part of the defence of the body against pathogens[135]. The genotype-phenotype association of several aGPCRs with inflammatory disease-states such as joint/bone inflammation or rheumatoid arthritis implies the potential of this receptor class for further immune disease contribution and treatment[136].

Cardiovascular diseases. Although many aGPCRs have been implicated in cardiac development, remodelling and vascular homeostasis in experimental systems, only a subset shows clear relevance in human cohorts. ADGRG1/GPR56 is the most robustly validated example. Studies using isolated human platelets show that ADGRG1/GPR56 functions as a collagen-binding receptor that mediates platelet activation and aggregation; pharmacological modulation with receptor agonists or antagonists alters platelet reactivity ex vivo, underscoring mechanistic relevance[71]. Plasma concentrations of the ADGRG1/GPR56 NTF are significantly higher within the first 24 h after ST-segment elevation myocardial infarction than at day 7, suggesting a role in acute thrombotic events in vivo[137]. Together, these findings support a direct link between ADGRG1/GPR56 activity and human thrombosis and myocardial infarction pathophysiology. To date, no aGPCR has been firmly associated with heart failure outcomes in human cohort studies. ADGRL4/ELTD1 remains a compelling candidate, as its expression is markedly reduced in hypertrophied ventricles of failing human hearts[68].

Metabolic diseases. In metabolic diseases, much of the mechanistic foundation derives from preclinical systems. Nevertheless, several aGPCRs now have human evidence linking them to obesity, type 2 diabetes/insulin resistance and non-alcoholic fatty liver disease (NAFLD). ADGRF5/GPR116 has emerged as a candidate regulator via interaction with the hepatokine/adipokine FNDC4. In patients after bariatric surgery, circulating *FNDC4* and *ADGRF5/GPR116* expression in visceral adipose tissue are increased[74]. In visceral adipose tissue from individuals with obesity, with or without impaired glucose tolerance, expression of *FNDC4* and/or *ADGRF5/GPR116* is also elevated[138]. Whether bariatric surgery further augments these obesity-associated increases remains to be determined. Collectively, these studies implicate this receptor–ligand pair in insulin resistance and dysregulation of glycaemic control in humans. In addition, a 2024 study has reported partially inactivating *ADGRL1/LPHN1* variants in an individual with obesity and identified additional rare variants in a paediatric adipose tissue cohort[78]. Beyond these examples, genotype–phenotype associations link multiple aGPCRs to indicators of metabolic dysfunction, including body weight, body fat, waist-to-hip ratio and altered concentrations of lipids, apolipoprotein B, glucose, leptin, insulin and adiponectin, supporting their relevance to human metabolism and prioritizing candidates for clinical interrogation[136].

Human cohort data implicate ADGRF1/GPR110 in steatogenic programmes. In NAFLD, hepatic *ADGRF1/GPR110* expression correlates with liver fat content, and mechanistic data identify stearoyl-CoA desaturase 1 (SCD1) as a downstream effector that promotes de novo lipogenesis. Suppression of ADGRF1/GPR110 or SCD1 mitigates steatosis in preclinical models, aligning with the correlative human observations and nominating ADGRF1/GPR110 as a potential target in human NAFLD[80].

Several aGPCRs are also implicated in human kidney disease. Altered expression of multiple aGPCRs has been reported across chronic kidney disease, diabetic nephropathy, lupus nephritis and focal segmental glomerulosclerosis, as well as renal cell carcinoma (summarized in ref. 139).

Bone and connective tissue disease. In bone and connective tissue disorders, ADGRG6/GPR126 and ADGRD1/GPR133 are particularly notable. Common variation in *ADGRG6/GPR126*, including SNPs, is associated with increased susceptibility to adolescent idiopathic scoliosis (AIS) across several populations[140]. Intronic variants in *ADGRG6/GPR126* have also been linked to scoliosis in individuals with Marfan syndrome[141]. In addition, rare *ADGRG6/GPR126* variants have been implicated in arthrogryposis multiplex congenita[19] and in aggressive periodontitis[142]. Common variation at this locus is further associated with human height, supporting a role in skeletal development and connective tissue biology[143]. Similarly, *ADGRD1/GPR133* has been associated with bone mineral density[144] and adult height through genome-wide association studies[145,146].

Infertility and pregnancy-related disorders. Pathogenic variants in *ADGRG2/GPR64* are linked to congenital bilateral absence of the vas deferens, a cause of male infertility owing to obstructive azoospermia[147]. Altered placental expression of *ADGRA2/GPR124*, *ADGRG6/GPR126* and *AGDGRE5/CD97* has been reported in cohorts with pre-eclampsia, a hypertensive disorder of pregnancy. Correlations with clinical phenotypes have been observed, although underlying mechanisms remain to be fully defined (summarized in ref. 148).

Cancer. Altered expression of aGPCRs is observed across several of the most common cancers worldwide (Global Cancer Observatory), including breast, colorectal, lung, prostate, thyroid and gastric cancers. There is increasing evidence that aGPCRs contribute functionally to multiple tumour types, although the frequency and direction of dysregulation vary by receptor and cohort. Across biopsy series and public datasets, all aGPCR subfamilies show differential expression in human tumours; expression of receptors such as *ADGRG1/GPR56* and *ADGRE5/CD97* are among the more frequently altered, whereas *ADGRA1-3/GPR123-125* and *ADGRL1-4/LPHN1-3,ELTD1* appear less commonly dysregulated. Notably, cancer cell lines often display a broader expression spectrum than primary tumours. The therapeutic and diagnostic potential of this receptor class is substantial: large extracellular N termini and tissue-specific splice isoforms may enable tumour-selective targeting with antibodies or nanobodies. For many aGPCRs, current knowledge is based on observed dysregulation, whereas mechanistic contributions to cancer biology are supported in selected cases by cell culture and mouse studies. Several reviews provide detailed discussion of mechanisms and exemplars across tumour growth, invasion, metastasis and angiogenesis[149,150].

Activation mechanisms of aGPCRs as basis for specific targeting

The complex architecture of aGPCRs supports an unusually diverse set of activation mechanisms that encompass both *cis* (Fig. 2a) and *trans* (Fig. 2b) activation. *Cis* signalling denotes receptor activation events that originate and are transduced within the same cell expressing the aGPCR. By contrast, *trans* activation refers to effects initiated through physical or functional interaction of an aGPCR with other receptors, either on the same cell or on neighbouring cells, wherein the observed (intra)cellular response is driven by the interactor rather than the index receptor itself.

Cis signalling mediated by the 7TMD

As in other GPCR classes, aGPCRs can bind small molecules and peptides within the 7TMD, thereby modulating their capacity to induce

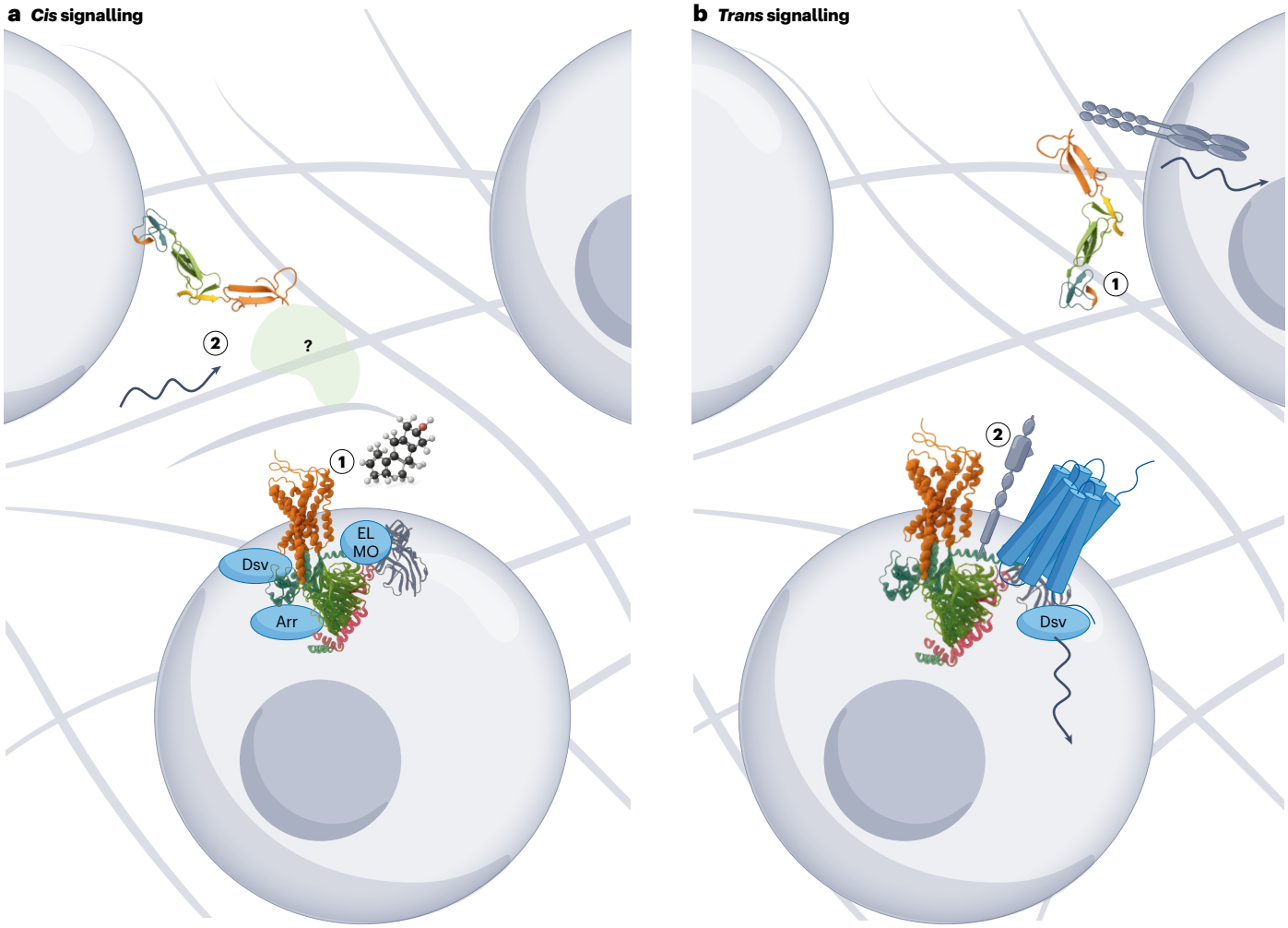


Fig. 2 | Activation mechanisms and signalling of adhesion GPCRs in physiological contexts. a, *Cis* signalling: In their active state, adhesion G-protein-coupled receptor (GPCR) seven-transmembrane domains (7TMD) engage transducers and adaptors, including heterotrimeric G proteins (example complex, PDB: 7EQ1), arrestins, dishevelled (DVL) or ELMO. Activation can arise (1) from endogenous orthosteric ligands (for example, steroid hormones or the Stachel peptides within the 7TMD) or (2) via N-terminal events that potentially expose the Stachel sequence. The extracellular N terminus integrates allosteric cues from extracellular matrix components, cell-surface ligands on neighbouring cells, and mechanical forces. **b**, *Trans* signalling: (1) Parts of the N terminus can act in an auto/paracrine fashion by binding receptors or channels on neighbouring or distant cells, whereas (2) lateral 7TMD-receptor complexes on the same cell modulate signalling.

downstream signalling. In 2012, a potential agonist for an aGPCR: beclomethasone dipropionate, was reported, which stimulated Gαo coupling to ADGRG3/GPR97 in a concentration-dependent manner, although no mechanistic or structural validation was available at that time[151]. Almost a decade later, beclomethasone dipropionate itself was found not to be the active ligand; rather, it undergoes hydrolysis to yield beclomethasone, which directly activates ADGRG3/GPR97 (ref. 152). Screening a series of cortisol-like compounds found many that could activate ADGRG3/GPR97. Importantly, cryo-EM structures captured cortisol and beclomethasone bound within the 7TM groove[152]. These studies, thus, represent a conclusive identification and structural validation of a physiologically relevant small-molecule ligand for an aGPCR. Building on these initial observations, several aGPCRs have since emerged as compelling candidates for steroid hormone and steroid-derivative binding. For example, dehydroepiandrosterone (DHEA), its sulfate (DHEAS) and deoxycorticosterone (DOC) occupy a ligand-binding pocket in the ADGRG2/GPR64 7TMD and induce Gs signalling[153]. 17α-Hydroxypregnenolone (17-OH PREG) acts as a weak ADGRG1/GPR56 agonist that triggers $G_{12/13}$ signalling[81]. Progestogen signalling further extends to ADGRG6/GPR126, for which progesterone and 17α-hydroxyprogesterone (17-OHP) activate the receptor via its CTF to induce Gi signalling[154]. In addition, 5α-dihydrotestosterone (DHT) and metandienone (MET) activate Gs/Go signalling of ADGRD1/GPR133 (ref. 89).

Beyond systemic activation by hormones, many aGPCRs harbour an agonistic sequence within their own polypeptide chain. Expression of the CTF alone confers constitutive activity in several aGPCRs[5,6], and mutational studies have identified an intramolecular agonistic Stachel sequence[2,3]. A recent study has profiled CTF constructs of all human aGPCRs across a selected panel of signalling

readouts, detecting activity for around 50% of receptors[155]. However, the absence of matched full-length aGPCR comparators, together with the omission of several canonical and non-canonical signalling modalities previously reported for aGPCRs, limits the inferences that can be drawn from the screen. In addition, the CTF constructs incorporated a maltose-binding protein–tobacco etch virus (MBP–TEV) fusion tag, which could impede Stachel–seven-transmembrane (7TM) interactions for some receptors, potentially explaining the failure to recapitulate activity for several constructs. Attempts to estimate the proportion of aGPCRs that genuinely harbour a functional Stachel sequence should also include testing of derived peptides to exclude artefactual effects of mutation, and a comprehensive assessment of canonical and non-canonical coupling to transducers and adaptors across species.

CTF structures across multiple aGPCRs reveal a highly similar Stachel-binding mode (reviewed in ref. 10). Transcriptome analyses show that variants encoding only the CTF exist[14]. However, most transcripts encode both NTF and CTF, which, on the basis of current structural data, would require release of the Stachel from the GAIN domain before engagement with the 7TMD. This release can occur as part of allosteric modulation mediated by the N terminus[156] or arise over time when NTF and CTF are no longer covalently attached[157,158].

Cis signalling mediated by the N terminus

The N terminus is the most prominent and complex structural feature of aGPCRs. Its hallmark adhesive modules and the GAIN domain integrate predominantly local cues, including interactions with extracellular matrix components, surface antigens and mechanical forces (Fig. 2). These inputs feed into an activation process that is still incompletely defined; in this context, the N terminus functions as an allosteric modulator of *cis* signalling.

A growing number of N-terminal ligands have been identified that activate aGPCRs (Table 2). For example, N-terminal binding of CD55 or LPAR1 to ADGRE5/CD97 activates G_{13}-dependent signalling[159]. Exposure of phosphatidylserine on the outer leaflet of HEK293T cells reduces SRE-driven luciferase signals through ADGRB1/BAI1 (ref. 160). ADGRD1/GPR133 responds to Plxdc2 (ref. 101), PTK7 (ref. 161) and N-terminally directed antibodies by elevating intracellular cAMP[162]. Several aGPCRs show context-dependent signalling: synaptamide binding to ADGRF1/GPR110 increases cAMP[163], whereas laminin-111 decreases it[164]. Likewise, different ligands of ADGRG6/GPR126, collagen IV[165], collagen VI[166] and laminin-211 (ref. 167), differentially modulate cAMP, with laminin-211 eliciting mechano-dependent effects[167]. ADGRG1/GPR56 is particularly pleiotropic: transglutaminase 2 (TG2)[168] and collagen III[169] activate RhoA-ROCK signalling, CD81 and CD9 promote receptor scaffolding with $G\alpha_{11}/G\alpha_q$[170], and laminin cooperates with TG2 to enhance SRE-driven transcription[171].

Although several ligand–receptor interactions have been functionally and structurally validated, the molecular transitions from binding to activation remain incompletely resolved. Mechanical stimuli, such as shear forces, vibration, shaking or cross-linking, have been implicated in N-terminally mediated activation of multiple aGPCRs (reviewed in refs. 172,173). A high-sensitivity mechanical stimulation assay combining a magnetic-tweezer system with a GPCR biosensor platform screened the mechanosensitivity of selected aGPCRs[60]. Using this method, five aGPCRs, ADGRD1/GPR133, ADGRG6/GPR126, ADGRL2/LPHN2, ADGRL3/LPHN3 and ADGRV1/VLGR1, triggered Gs or Gi signalling in response to force stimulation, corroborating prior reports of mechanosensitivity for these receptors[39,156,162,167,174,175].

A prevailing hypothesis posits that mechanical forces promote NTF shedding, thereby unmasking the constitutively active CTF. This model is consistent with several observations: (1) isolated CTFs show constitutive signalling[5,6]; (2) mechanical stimuli increase NTF shedding[71,159,176], and (3) GPS motif mutations diminish mechanosensitivity[171,177]. As a result, detectable NTF shedding is often equated with aGPCR activation. However, only a few studies measure intracellular signalling or cellular function alongside NTF shedding after mechanical challenge. A notable exception is ADGRE2/EMR2, for which vibration, with or without dermatan sulfate, induces mast cell degranulation, and mutations that increase NTF shedding exacerbate this response[22]. Additional support for NTF dissociation-dependent and Stachel-dependent activation comes from chimeric PAR1/Flag-enterokinase-CTF constructs that enable enzyme-induced removal of the N terminus, resulting in measurable increases in signalling[158,178].

There is also evidence that complete NTF dissociation is not required for activation. In cleavable ADGRE2/EMR2, NTF and CTF still co-localize after vibration, yet degranulation occurs; introducing the p.C^{492}Y mutation within the GAIN domain causes a marked loss of NTF at the cell surface in response to force and further increases degranulation[22]. Thus, NTF shedding is not a prerequisite for mechanical activation, but may represent an upper bound of this process. Moreover, some aGPCRs are mechanosensitive despite resistance to cleavage, either naturally[179] or owing to mutations[39,174]. Cleavage-suppressing mutations often produce divergent functional outcomes, in part because it is difficult to distinguish the effect of impaired cleavage from disruption of the Stachel sequence, which shares overlapping residues. One study has introduced two cleavage-disrupting mutations into *Drosophila* latrophilin/dCIRL at the GPS (H−2 L−1↓T + 1): an H → A change at the −2 position (dCirlH > A) and a T → A change at +1 (dCirlT > A)[180]. Only the T → A mutant, directly affecting the Stachel, abolished responses to mechanical stimuli. Other studies report variable impacts from mutating the H−2 position, which should affect cleavage but not the Stachel: in ADGRD1/GPR133, H^{543}R does not alter shake-induced activation[174], yet abrogates activation by an N-terminal antibody[162]. This variability reinforces that mechanical inputs on the N terminus can produce graded structural rearrangements and, consequently, graded activation states. Single-molecule magnetic-tweezer measurements on purified GAIN domains show that GAIN-Stachel dissociation occurs at low pico-newton forces, preceded by GAIN unfolding that may transiently expose the Stachel, allowing intermediate activation states[181,182]. Notably, these studies exclude the 7TMD and potential stabilization by endogenous ligands. Dynamic structural information integrating all receptor modules remains insufficient to fully explain the complexity of this activation mechanism. Recent studies suggest that the N terminus also modulates steroid sensing by aGPCRs as shown for the ADGRG2/GPR64-targeting nanobody Nb23-bi, which potentiates DHEA-induced receptor activation and rescues dysfunction of pathogenic ADGRG2/GPR64 missense variants associated with congenital absence of the vas deferens[183]. Structural analysis shows that Nb23-bi positions DHEA approximately 3.9 Å deeper within the orthosteric pocket and reduces its distance to the toggle-switch residue from 5.6 Å to 4.0 Å.

Cis signalling-independent roles of the N terminus

Several ligand-aGPCR pairs show robust binding without evidence, so far, of *cis* signalling induction. For ADGRE5/CD97, dermatan sulfate[184], $\alpha_5\beta_1/\alpha_v\beta_3$ integrins[185] and CD90 (ref. 186) were among the first ligands identified for any aGPCR, yet they have not produced

Table 2 | Adhesion GPCR endogenous ligand and physiological signification

Receptor	aGPCR endogenous ligand	Physiological significance	Cryo-EM structure	Refs.
ADGRA2/GPR124	$\alpha_v\beta_3$-integrin	Adhesion and migration during angiogenesis	n.a.	247
	WNT7a/b	Mammalian limb development	n.a.	248
	Glycosaminoglycans	CNS vascularization and BBB establishment	n.a.	199
ADGRB1/BAI1	$\alpha_v\beta_5$-integrin	Endothelial cell proliferation	n.a.	249
	Phosphatidylserine	Macrophage engulfment	n.a.	250
	Lipopolysaccharide	Macrophage engulfment	n.a.	251
	RTNR4	Neuronal development	pdb_00007r86	187
	CD36	Inhibition of angiogenesis	n.a.	198
ADGRB3/BAI3	C1ql1–C1ql4	Synapse formation	pdb_00008vwy	189,252–254
	C1q-like-3	Insulin secretion	n.a.	255
	C1qL4, Stabilin-2	Myoblast fusion	n.a.	256
ADGRC2/CELSR2	Homophilic interactions	Axon guidance; neurite growth	n.a.	257
	Cytosolic glycyl-tRNA synthetase (GARS1)	Cancer cell death	n.a.	258
ADGRC3/CELSR3	Homophilic interactions	Axon guidance; neurite growth	n.a.	257
	Dystroglycan	Axon guidance; neurite growth	n.a.	190
ADGRD1/GPR133	5α-DHT	Skeletal muscle strength	pdb_00009iv1	89
	PTK7	Glioblastoma growth, osteoblast differentiation	n.a.	90,161
ADGRE2/EMR2	Chondroitin sulfate	Adhesion	n.a.	184,259
ADGRE5/CD97	Chondroitin sulfate	T cell and B cell interaction	n.a.	184
	$a_v\beta_3$-integrin, $\alpha_5\beta_1$-integrin	Angiogenesis	n.a.	185
	LPA receptor	Tumour invasion	n.a.	260
	CD90	Leukocyte trafficking to inflammatory sites	n.a.	186
	CD55	T cell activation	pdb_00007do4	261–264
ADGRF1/GPR110	Synaptamide	Synaptogenesis	n.a.	163,233
	Occludin	BBB integrity	n.a.	265
ADGRF5/GPR116	FNDC4	Glucose homeostasis	n.a.	74
	Surfactant Protein-D	Pulmonary surfactant pool size regulation	n.a.	86,87
ADGRG1/GPR56	17α-Hydroxypregnenolone	Ferroptosis	n.a.	81
	Collagen III	Cortical development and lamination; haemostatic plug formation	n.a.	71,169,266
	Heparin	Cell adhesion and migration	n.a.	267
	Transglutaminase-2	CNS myelination, myotube hypertrophy, melanoma progression	n.a.	268–272
	Progastrin	Colonic mucosal proliferation and carcinogenesis	n.a.	273
	Phosphatidylserine	Synaptic pruning	n.a.	191
ADGRG2/GPR64	DHEA, DHEAS	Alleviate testicular inflammation	pdb_00007xke	153
ADGRG3/GPR97	Glucocorticoids	Adrenal cortex secretion	pdb_00007d76	152
	Astragaloside IV	Protection against hyperhomocysteine-exacerbated sepsis-associated acute kidney injury	n.a.	274
	L-Phe	Presumably produced by gut microbiota to induce Go signalling	n.a.	275
ADGRG6/GPR126	Progesterone	Breast cancer development	n.a.	154
	17-Hydroxyprogesterone	Breast cancer development	n.a.	154
	Collagen IV	Peripheral nerve development	n.a.	165
	Laminin-211	Schwann cell development	n.a.	167
	Collagen VI	unknown	n.a.	166
	Cellular Prion Protein	Schwann cell function	n.a.	232

Table 2 (continued) | Adhesion GPCR endogenous ligand and physiological signification

Receptor	aGPCR endogenous ligand	Physiological significance	Cryo-EM structure	Refs.
ADGRL1/LPHN1	TAFA2	Suppresses apoptosis	n.a.	276
	α-Latrotoxin	Toxin docking with cells	n.a.	277–279
	Teneurins	Neuronal pathfinding and synaptogenesis	pdb_00006ska	280–284
	Neurexins	Transsynaptic connection formation	n.a.	285,286
	FLRT proteins	Synaptic development	n.a.	284,287
ADGRL2/LPHN2	PCLAF	Bone marrow adipocyte senescence	n.a.	288
	Teneurins	Axon guidance	pdb_00006ske	280
	FLRT proteins	Synaptic development	n.a.	287
ADGRL3/LPHN3	Teneurins	Neuronal reshaping; synapse formation; axon guidance	pdb_00005ftt	280,289
	FLRT proteins	Synaptic development	pdb_00005cmn	284,289,290

aGPCR, adhesion G-protein-coupled receptor; BBB, blood–brain barrier; CNS, central nervous system; cryo-EM, cryogenic electron microscopy; n.a. not available.

measurable downstream signalling. Similar observations exist for other aGPCRs, including ADGRB1/BAI1 with RTN4 (ref. 187) and neuroligin-1 (ref. 188); ADGRB3/BAI3 with C1q-like proteins[189]; ADGRC3/CELSR3 with dystroglycan[190]; ADGRE2/EMR2 with dermatan sulfate[132]; ADGRF5/GPR116 with surfactant protein D[86]; ADGRG1/GPR56 with phosphatidylserine[191]; ADGRL1/LPHN1 with contactin-6 (ref. 192); and ADGRL2-ADGRL3/LPHN2-LPHN3 with teneurin-2–4 (refs. 193,194). It remains possible that these interactions have not yet been probed with the appropriate signalling readouts. Equally, some ligands might primarily support adhesion or maintain full-length receptor context to prevent NTF diffusion after spontaneous or induced separation from the CTF. Finally, NTF–ligand interactions may serve *trans* signalling roles.

Recipients of N-terminally mediated *trans* activation can be direct adhesion partners or receptors positioned at distant anatomical sites (Fig. 2). Many examples imply N-terminal functions independent of *cis* signalling; here we focus on cases with functional readouts and refer to recent reviews for broader coverage[8,195]. The entire NTF of ADGRB1/BAI1 (Vstat-120) and a shorter fragment (Vstat-40), generated by MMP14, can be released to act as angiogenesis inhibitors[196,197]. These anti-angiogenic effects are mediated primarily by CD36 on endothelial cells, with Vstat-120 also binding and blocking $\alpha_v\beta_5$ integrins[198]. Similarly, the NTF of ADGRA2/GPR124 can be released intact and further processed into smaller fragments by an unidentified protease; fragments containing the hormone receptor domain bind integrins and glycosaminoglycans and promote endothelial cell survival[199]. The EGF-like repeats of the ADGRE5/CD97 NTF serve as chemoattractants for endothelial cells and contribute to angiogenesis in vitro and in vivo via interactions with integrins $\alpha_5\beta_1$ and $\alpha_v\beta_3$, a mechanism particularly relevant in tumourigenesis[185]. A distinct NTF-only function exists for ADGRG3/GPR97, which activates neutrophils in vitro by engaging membrane proteinase 3 and thereby proteolytically activating PAR2; this activity requires a multiprotein complex that includes CD177 and CD16b[200].

In the nervous system, the N terminus of ADGRB1/BAI1 functionally interacts with neuroligin-1 to promote spine growth and synaptogenesis[188]. It also binds reticulon-4 receptors (RTN4Rs; also known as Nogo receptors) via a single thrombospondin type-1 repeat, a critical interaction for driving synapse formation and excitatory transmission through RTN4RL2 (ref. 187). During development, sequential expression of ADGRB1/BAI1 in neurons and glia is thought to fine-tune synaptogenesis via binding to neuronally expressed RTN4Rs. ADGRL3/LPHN3 provides another example by regulating glutamate release from cone cells through negative modulation of the voltage-gated Ca^{2+} channel Cav1.4 (ref. 201). This process, crucial for daytime visual processing, is governed by alternative splicing: exon 5 encodes an N-terminal DN domain that reduces binding to the partners teneurin and FLRT. In horizontal cells, an alternatively spliced variant lacking exon 5 is highly expressed, enabling stronger interactions with cone-cell ligands at lower concentrations, thereby inhibiting Cav1.4 and modulating glutamate release.

In *C. elegans* the N terminus of latrophilin-1 (LAT-1) is required to maintain fertility[30,202]. Mechanistically, the GAIN and RBL domains of LAT-1 bind the Notch ligand LAG-2 on the same cell to promote Notch receptor signalling in the opposing cell[29].

Trans signalling mediated by the 7TMD and C terminus

Functional complex formation is not restricted to the N terminus; the 7TMD and C terminus also participate in protein–protein interactions that regulate other receptors. The best-studied example is participation in Wnt signalling complexes, an evolutionarily conserved pathway central to development, physiology and disease. In the neurovascular unit, cerebral endothelial cells must express an ADGRA2/GPR124–RECK membrane complex to respond to neural-progenitor-derived WNT7 ligands and activate canonical Wnt/β-catenin signalling[203,204]. Notably, this scaffolding role of ADGRA2/GPR124 does not depend on classical GPCR activity; instead, it relies on extracellular binding to RECK and intracellular recruitment of DVL[205]. This assembly locally concentrates ligands for Frizzled activation and downstream signal propagation. ADGRA3/GPR125, a close homologue, also interfaces with Wnt signalling by interacting with DVL1 and the receptor tyrosine kinase RYK, a Wnt receptor involved in neuronal differentiation and axon guidance[50,206].

Members of the cadherin EGF LAG seven-pass G-type receptor family (ADGRC1–ADGRC3/CELSR1–CELSR3) are similarly implicated in Wnt pathways[98,207]. All three ADGRC/CELSR proteins interact with RYK, and ADGRC1/CELSR1 forms additional signalling complexes[208]. These complexes localize to adherens junctions and mediate RHO-dependent actomyosin contractility, contributing to cytoskeletal remodelling. ADGRC1/CELSR1 also binds VANGL2, a core planar cell polarity protein, forming a partnership essential for processes such as hair follicle orientation[209].

Signalling diversity

The complexity of aGPCR function is mirrored by the diversity of intracellular *cis* signals mediated through the 7TMD and C terminus via distinct transducers and adaptor proteins (Fig. 2). Many aGPCRs couple to multiple classes of heterotrimeric G proteins (Gs, Gq, Gi/o and G12/13), to β-arrestins, and to cytoplasmic adaptor proteins including PDZ-domain-containing proteins, ELMO1, DVL and DAAM1. Downstream outputs are often receptor specific and context dependent, reflecting the intricate regulatory architecture of the family.

G protein coupling is frequently promiscuous and stimulus dependent. For example, ADGRD1/GPR133, when activated by 5α-DHT, couples to Gs and G12, whereas activation by the androgen derivative MET promotes coupling to Gs and Go[89]. 17-OH PREG stimulates ADGRG6/GPR126 to activate Gi at low concentrations, but triggers Gs at higher concentrations[154]. In addition to steroid-induced signals, the Stachel can elicit selective, context-dependent transduction in full-length receptors engaged by N-terminal ligands. Conversely, truncated CTFs can represent an active state that enables broadly promiscuous G protein coupling[155,210] or induces signalling switches[135]. Stachel-derived peptides can likewise initiate multiple pathways via promiscuous G protein engagement[3,211].

Beyond G proteins, several aGPCRs recruit β-arrestins to mediate G protein-independent or scaffolded signalling. ADGRG2/GPR64 recruits β-arrestin-1 as a scaffold to regulate CFTR channel activity, which is essential for luminal fluid reabsorption in the male reproductive tract[100]. Recruitment of β-arrestin-2 to MBP–CTF fusion constructs has been reported for ADGRE1/EMR1, ADGRE3/EMR3, ADGRF1/GPR110, ADGRF4/GPR115 and ADGRG7/GPR128 in BRET assays[155].

In parallel, aGPCRs interact with a variety of cytoplasmic adaptors that fine-tune downstream signalling. ADGRB1/BAI1, ADGRB3/BAI3 and ADGRG7/GPR128 bind ELMO1, a regulator of cell migration and cytoskeletal dynamics[212–214]. ADGRA3/GPR125 interacts with DVL in zebrafish, facilitating recruitment of planar cell polarity components[50]. ADGRV1/VLGR1 binds multiple PDZ-domain proteins, including PDZD7 and WHRN, to regulate canonical and non-canonical signalling; notably, PDZD7 negatively regulates Gi coupling by ADGRV1/VLGR1, indicating interplay between G protein and non-canonical pathways[215].

Current state and future perspectives of aGPCRs as therapeutic targets

Given the extensive pathophysiological footprint of aGPCRs, there is strong interest in identifying activating and inhibiting compounds (current modulators are listed in Tables 3–5; structures of small-molecule modulators may be found in Fig. 3). Specific agonists, antagonists and inverse agonists enable direct interrogation of receptor function in vitro and in vivo, complementing KO studies that can be confounded by compensatory cellular mechanisms. Such modulators also provide a foundation for drug development to exploit this receptor class therapeutically. Targeting strategies currently focus on the N terminus and the orthosteric 7TM pocket (Fig. 4a).

Antibodies and nanobodies with therapeutic potential

Antibodies and nanobodies are emerging as promising therapeutic tools owing to their potential for targeted interactions with specific receptor and transcript variants. The N terminus of aGPCRs is particularly promising for such targeted therapies. Transcript variations in aGPCRs are often tissue-specific and cancer-specific, affecting both basal and induced signalling capacities[14,15]. Antibodies and nanobodies can modulate signalling through various mechanisms: inducing conformational changes, cross-linking or blocking endogenous ligand interactions. Multiple in vitro and in vivo studies have demonstrated the feasibility of antibody and nanobody-based modulation (Table 3). For example, targeting the ADGRE2/EMR2 N terminus with a polyclonal antibody (AF4894) has been shown to induce NFAT-mediated luciferase production in full-length receptors[216]. Similarly, a monoclonal antibody (2A1) enhances LPS-induced IL-1β production in monocyte cell cultures by activating PLC-β signalling[217]. A synthetic antibody fragment, LK30, targets the Lec domain of ADGRL3/LPHN3 to mediate two functions in vitro: it induces SRE-signalling and blocks the N-terminal interaction with teneurin 2, thereby disrupting cell–cell adhesion[218]. SRE signalling

Table 3 | Antibodies and nanobodies modulating aGPCR function

Receptor	Antibody	Antibody or nanobody	Physiological significance	Compound developmental stage	Refs.
ADGRE1/EMR1	C1E7	Antibody	NK cell-mediated killing of eosinophils	Preclinical — cynomolgus macaque	226
ADGRE2/EMR2	2A1, AF4894	Antibodies	Gα16-dependent PLC-β activation, NFAT luciferase reporter	In vitro testing	216,217
ADGRE5/CD97	1B2	Antibody	Delta arthritis score reduction, joint damage protection	Preclinical — mouse model	133,225
ADGRG1/GPR56	4C3, 11E9	Antibodies	Enhance migration of U87-MG cells through G_q/Rho	In vitro testing	219
	CG4	Antibody	G12/13 signalling to induce IL-6-mediated migration and invasion of melanoma cells	In vitro testing	220
	10C7	Antibody	Recruitment of β-arrestin-2, G13/Rho signalling, antitumour function as ADC	Preclinical — mouse model	222–224
ADGRG2/GPR64	Nb23-bi	Nanobody	Promotes DHEA-induced activation and reverses mutant-induced dysfunction	Preclinical — mouse model	183
ADGRL3/LPHN3	rhLPHN3	Antibody	Reduction of clinical signs of asthma and inflammation markers	Clinical — paediatric asthma	227
	LK30[a]	Synthetic antibody fragment	Competes with teneurin binding	In vitro testing	218

ADC, antibody–drug conjugate; aGPCR, adhesion G-protein-coupled receptor. [a]Cryogenic electron microscopy structure available of ADGRL3-lectin domain in complex with LLK30 (pdb_00008djg).

Table 4 | Synthetic aGPCR peptide ligands

Receptor	aGPCR peptide ligand (potency)	Physiological significance	Sequence	Refs.
ADGRB1/BAI1	Stachel peptide (p19, EC_{50} n.d.)	Synapse formation, spine density	STFAILAQL SADANMEKAT	188
ADGRD1/GPR133	Stachel peptide (p13, EC_{50} > 400 μM)	cAMP increase	TNFAILMQVVPLE	2,174
ADGRF1/GPR110	Stachel peptide (p12, EC_{50} = 37 μM in Gq GTPγS; p13, EC_{50} = 167.2 μM in cAMP and EC_{50} = 19.6 μM in IP_3 assay)	Gq GTPγS binding, cAMP and IP3 increase	TSFSILMSPFVP/S	3,211
ADGRF5/GPR116	Stachel peptide (p13, EC_{50} = 75.8 μM in IP_3 and EC_{50} = 25.39 μM in calcium assay)	IP_3/calcium increase/ß-arrestin interaction, maintain the skeletal muscle stem cell pool, suppression of surfactant phospholipid secretion in mouse	TSFSILMSPDSPD	177, 211,291
ADGRG1/GPR56	Stachel peptide (p7, EC_{50} = 35 μM)	G13 GTPγS binding, treatment of depression in mouse	TYFAVLM	3,117
ADGRG2/GPR64	Stachel peptide (p15, EC_{50} = 77 μM)	cAMP increase	TSFGILLDLSRTSLP	228,292
	Optimized Stachel peptide (VPM-p15, EC_{50} = 1.4 μM	cAMP increase	T1V/F3(1-Nal) and T1V/F3Phe(4-Me)-p15	229
ADGRG5/GPR114	Stachel peptide (p7, EC_{50} n.d.; p18, EC_{50} > 100 μM)	CRE/cAMP increase	TYFAVLMQ/ LSGDPVPAELQV	174, 179,293
ADGRG6/GPR126	Stachel peptide (p16, EC_{50} > 400 μM; p16G4A, EC_{50} n.d.)	cAMP increase or decrease, Schwann cell myelination rescue in *Danio rerio*	THFG(A) VLMDLPRSASQL	2
	Prion protein flexible tail peptide ($FT_{23\text{-}50}$: EC_{50} = 850 nM)	cAMP increase in primary Schwann cell culture, Schwann cell myelination rescue in *D. rerio*	GWNTGGSRYPG QGSPGGNRYP	232
ADGRL1/LPHN1	Teneurin C-terminal-associated peptide (TCAP-1, EC_{50} n.d.)	Modulation of cytoskeletal organization and intercellular adhesion and aggregation; increase of testis size and attenuation of stress, as well as morphine withdrawal symptoms in mice	QQLLGTGRVQ GYFVLSVEQYLEL SDSANNIHFMRQSEI	230,231
	Stachel peptide (p12, EC_{50} n.d.)	cAMP increase, lethality and tissue polarity rescue in *Caenorhabditis elegans*	THFAVLMDVRGH	28

aGPCR, adhesion G-protein-coupled receptor; EC_{50}, half-maximal effective concentration; IP_3, inositol trisphosphate; n.d., not determined.

is abolished within the $T^{842}G$ mutant, which disrupts cleavage but might also affect the Stachel sequence. Thus, the precise cause of the signalling deficit cannot be determined. For ADGRG1/GPR56, multiple antibody modalities have revealed distinct signalling outcomes. Monoclonal antibodies raised against the N terminus induce migration of human glioma U887-MG cells via a Gq/Rho pathway[219]. An immobilized polyclonal N-terminal antibody (CG4) drives $G\alpha_{12/13}$/RhoA-dependent IL-6 secretion in melanoma cells, promoting migration and invasion[220]. CG4-mediated activation depends on the co-expression of CD9 and CD81 and on a functional Stachel sequence, with positions 3, 6 and 7 exerting the greatest impact on signalling. The ADGRG1/GPR56 $T^{383}A$ mutant shows markedly reduced CG4-mediated IL-6 secretion — interpreted as an impact of GPS cleavage on receptor function — and CG4 stimulates NTF release in a CD9/CD81-dependent manner. However, because $T^{383}A$ also perturbs both the intramolecular agonist sequence[3] and GPS cleavage[221], it remains unclear which domain change drives the reduced activity. The antibody 10C7, which targets the GAIN domain of ADGRG1/GPR56, induces $G_{12/13}$-RhoA signalling and Src-Fak adhesion signalling in HEK293T cells[222,223]. This activation requires a PLL domain upstream of the GAIN domain. The $T^{383}A$ mutant exhibits significantly reduced, but not abolished, SRE response element induction[222] and retains WT-like $G_{12/13}$ recruitment[223] despite similar surface expression. Notably, 10C7 co-internalizes with ADGRG1/GPR56, enabling an antibody–drug conjugate strategy in which 10C7 is coupled to the alkylating agent DMSA. This construct achieved IC_{50} values of 93.6–696 pmol l^{-1} in cell viability assays and significantly reduced tumour volume at 5 mg kg^{-1} across several cancer cell lines in xenograft models[224]. In ADGRG2/GPR64, the nanobody Nb23 bi recognizes the NTF and increases DHEA potency for Gs and Gq signalling[183]. Cryo-EM structures reveal that Nb23 bi shifts DHEA deeper into the 7TMD binding pocket, and functional relevance for this combined activation was demonstrated in an LPS-induced orchitis model, wherein Nb23 bi synergized with DHEA to enhance anti-inflammatory effects. For ADGRE5/CD97, the monoclonal antibody 1B2 ameliorates joint inflammation in experimental arthritis and confers protection when administered before symptom onset[133,225]. The anti-inflammatory effect is attributed to impaired ADGRE5–CD55 interaction, which reduces immune cell recruitment. Because 1B2 binds EGF domain 1 and recognizes both full-length receptor and shed NTF, the antibody is cleared from circulation and accumulates in organs with high receptor expression. Clinically relevant is the finding that arthritis increases the levels of shed NTF, necessitating higher 1B2 doses to achieve the desired effects. In cynomolgus monkeys, a single 5 mg kg^{-1} intravenous dose of an afucosylated anti-ADGRE1/EMR1 antibody enhanced NK cell-mediated depletion of eosinophils, suggesting a potential treatment strategy in eosinophilia, wherein ADGRE1/EMR1 expression positively correlates with the disease[226].

Clinically, an antibody targeting ADGRL3/LPHN3 has been evaluated in paediatric asthma[227]. In this study, 342 children aged 4–11 years old with capillary asthma and bronchitis were randomized to receive salbutamol (12 mg day^{-1}), montelukast (12 mg day^{-1}) or a recombinant anti-ADGRL3/LPHN3 antibody (rhLPHN3 Ab; 12 mg day^{-1}) by nasal aerosolization for 16 weeks. Higher antibody concentrations were

not well tolerated, with treatment-related adverse events including hypertension, proteinuria, fatigue, hypertriglyceridemia, constipation and oedema; even at 12 mg day^{-1}, hypertension and proteinuria occurred in more than 10% of patients. Nonetheless, over 16 weeks, rhLPHN3 outperformed salbutamol and montelukast with respect to onset frequency and duration of asthma attacks, duration of capillary bronchitis, and number of nights with sleep disruption. These phenotypic improvements correlated with favourable biomarker changes, such as reduced PKA phosphorylation, diminished NFκB activation, and lower expression of E-selectin and MUC5AC in airway smooth muscle cells, alongside significantly increased proliferation and adhesion indices in these cells.

Together, these examples underscore both the promise and complexity of antibody and nanobody therapeutics targeting aGPCRs. Mechanistically, mutations such as ADGRG1/GPR56 $T^{383}A$ affect both GPS cleavage and the intramolecular agonist sequence, complicating causal attribution and highlighting the need for careful construct design and interpretation. Context matters: co-receptors such as CD9/CD81 and dynamic NTF–CTF interactions can decisively shape antibody effects, including NTF shedding and downstream pathway selection. Translation to the clinic remains encouraging but nuanced; although targeted activation or blockade can be potent, tolerability (as seen with rhLPHN3 adverse effects) and target-dependent tissue sequestration or antigen sink (as seen with 1B2) are practical hurdles that require proactive management. At the same time, opportunities are clear: antibodies that co-internalize with their targets (for example, 10C7) are strong antibody–drug conjugate candidates, and nanobodies can allosterically tune small-molecule ligand binding to amplify therapeutic effects (for example, Nb23 bi with DHEA).

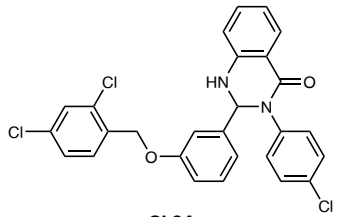
Synthetic peptide ligands

Mimicking the intramolecular agonistic Stachel sequence with synthetic peptides is an attractive strategy for targeting aGPCRs. These derived peptides engage the orthosteric pocket within the 7TMD, which overlaps the cavity used by the endogenous intramolecular agonist[228]. Early studies have revealed modest potency, with half-maximal effective concentration (EC_{50}) values in the double-digit to triple-digit micromolar range[2,3]. Nevertheless, a first ADGRG6/GPR126-derived peptide successfully rescued a hypomorphic myelination defect in zebrafish[2], and initial mutagenesis already yielded an antagonistic peptide for ADGRG6/GPR126 (ref. 2). Soon thereafter, several additional agonistic Stachel peptides were identified across the aGPCR family, bringing the total to nine receptors responsive to this mode of stimulation (Table 4). Four Stachel peptides have been used to modulate aGPCR function in vivo: beyond the zebrafish model, lethality in heterozygous-deficient ADGRL1/Lat-1 nematodes can be rescued[28]; in mice, ADGRF5/GPR116-derived Stachel peptides modulate surfactant production[177]; and an ADGRG1/GPR56-derived agonistic peptide has been used to treat depression in mice[117]. That said, the modest efficacy and limited specificity of current Stachel peptides constrain their druggability[9,211]. Owing to high sequence conservation, peptides derived from one aGPCR can activate several others, both within and across subclasses[211]. To address these limitations, systematic alanine-scanning alongside natural and noncanonical amino acid substitutions enabled development of an improved ADGRG2/GPR64 Stachel-derived agonist (VPM p15) with more than 100-fold higher affinity than the WT peptide[229]. Guided by available cryo-EM structures, the study has pinpointed the toggle switch residue $W^{6.53}$ and ECL2 residues of ADGRG2/GPR64 as key determinants of VPM p15 binding. To generate antagonists for

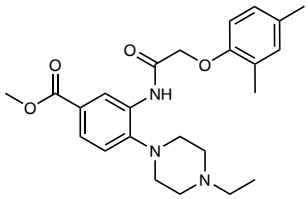
Table 5 | Small-molecule modulators of aGPCRs

Receptor	Ligand (potency)[a]	Physiological significance	Compound developmental stage	Refs.
ADGRD1/GPR133	GL64 (EC_{50} = 3.98 μM on full-length receptor)	Inhibits osteoclast function through Gαs activation	Preclinical — mouse model	91
	AP-970/43482503 (AP503, EC_{50} = 1.2 nM on full-length receptor)	Enhances osteoblast function and differentiation through Gαs activation, cAMP-mediated ß-catenin activation	Preclinical — mouse model	89,90
ADGRF1/GPR110	*N*-Docosahexaenoylethanolamine (synaptamide, EC_{50} = 4.1 nM in cultured cortical neurons); dimethylsynaptamide (A8)	Neuroprotection and development, anti-inflammation, inhibits osteoclast function, Gs-cAMP signalling	Preclinical — mouse model	163,233,236,294
ADGRG1/GPR56	3-α-Acetoxydihydrodeoxygedunin (α-DOG, EC_{50} = 4.8 μM on partially truncated Stachel mutant)	Partial agonist, enhances G13 activation, requires partial Stachel and cleavage	In vitro testing	171,240
	Dihydromunduletone (DHM, IC_{50} = 20.9 μM on CTF mutant)	Antagonist	In vitro testing	241
	2-(furan-2-yl)-1-[(4-phenylphenyl)carbonyl]pyrrolidine (compound **4**, EC_{50} n.d.) Propan-2-yl-4-(2-bromophenyl)-2,7,7-trimethyl-5-oxo-1,4,5,6,7,8-hexahydroquinoline-3-carboxylate (compound 36, EC_{50} = 2.95 μM on Stachel-deficient truncation mutant)	Compound 36 agonist and compound **4** partial agonist, enhance G13 activation	In vitro testing	242
ADGRG3/GPR97	Beclomethasone dipropionate (EC_{50} = 3.1 nM on full-length receptor)	Go signalling, attenuates psoriasis-like symptoms	Preclinical — mouse model	66,151,152
	Curcumin (EC_{50} n.d.)	Activation of SRE-mediated luciferase	In vitro testing	239
ADGRG5/GPR114	α-DOG (EC_{50} n.d.)	Partial agonist, enhances GTPγS	In vitro testing	240
	Dihydromunduletone (DHM, EC_{50} n.d.)	Antagonist, decreases G13 signalling	In vitro testing	241
ADGRG6/GPR126	Apomorphine hydrochloride (EC_{50} = 21.5 μM on full-length receptor)	Promotes Schwann cell development	Preclinical — zebrafish model	25

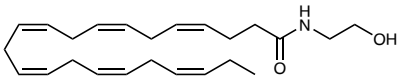
aGPCR, adhesion G-protein-coupled receptor; CTF, C-terminal fragment; EC_{50}, half-maximal effective concentration; n.d. not determined. [a]See Fig. 3 for ligand structures.



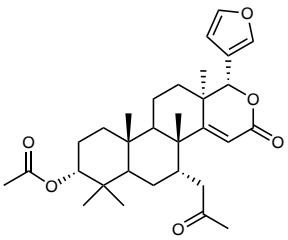
GL64



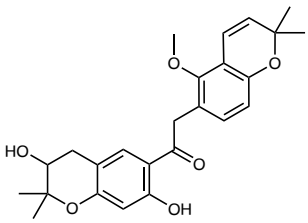
AP503



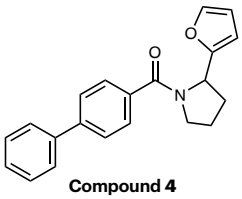
N-Docosahexaenoylethanolamine



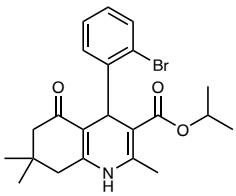
3-α-Acetoxydihydrodeoxygedunin



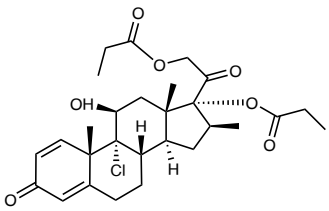
Dihydromunduletone



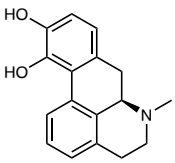
Compound 4



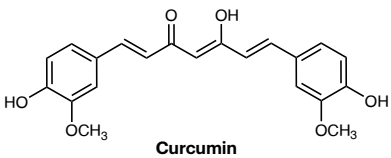
Compound 36



Beclomethasone dipropionate



Apomorphine



Curcumin

Fig. 3 | Small-molecule modulators of aGPCRs. Chemical structures of small-molecule compounds shown to activate adhesion G-protein-coupled receptors (aGPCRs) (as listed in Table 5).

ADGRG2/GPR64, targeted mutations within the Stachel sequence (F601D/E) produced peptidic inhibitors that abolished receptor signalling while retaining high-affinity binding — a strategy successfully extended to three additional aGPCRs: ADGRG4/GPR112, ADGRG5/GPR114 and ADGRG6/GPR126 (ref. 228).

Because some endogenous aGPCR ligands are proteins, it is also plausible that short peptide fragments derived from these proteins can act as activators. This has been demonstrated for the ADGRL1/LPHN1 ligand teneurin C-terminal-associated peptide (TCAP-1), which modulates adhesion and cytoskeletal organization in vitro[230]. TCAP-1 has also been used in vivo to modulate testis function and to alter stress phenotypes and morphine withdrawal symptoms in mice[231]. In a related example, a peptide derived from the flexible tail of the prion protein rescues myelination defects in Adgrg6/Gpr126-hypomorph zebrafish[232], mirroring observations with the Stachel peptide[167]. For both of these non-Stachel peptides, however, the precise binding interfaces remain unknown, underscoring a key gap for mechanistic mapping and rational optimization.

Small-molecule compounds

N-Docosahexaenoylethanolamine (synaptamide), the endogenous ligand of ADGRF1/GPR110, can also be deployed therapeutically. Synaptamide induces neurite outgrowth in a cAMP-dependent manner, and pull-down experiments with biotinylated synaptamide identified ADGRF1/GPR110 as its binding partner, with low-nanomolar potency in cultured neurons[163]. Synaptamide-mediated activation is neither cleavage-dependent nor Stachel-dependent but requires the NTF, and in-cell chemical cross-linking suggests binding within the GAIN domain[233]. Docking of synaptamide into the crystal structure of the ADGRF1/GPR110 N terminus supports this view, predicting binding in the region of the Stachel peptide[234]. The working model is that synaptamide occupancy displaces the Stachel, enabling its engagement with the 7TMD pocket. Intravitreal injection of synaptamide following optic nerve crush injury supports axon extension in WT but not ADGRF1/GPR110-KO mice[235]. Intraperitoneal application of synaptamide (5 mg kg^{-1}) modulates ADGRF1/GPR110 in mouse macrophages to reduce inflammatory mediator production after lipopolysaccharide challenge[236] or acute kidney injury[237], and in cell culture, it inhibits osteoclastogenesis in a ADGRF1/GPR110-dependent fashion[238].

Steroid hormones are endogenous soluble agonists that bind the 7TMD of aGPCRs. Accordingly, beclomethasone dipropionate was identified as the first small-molecule agonist, with an EC_{50} of 3.1 nM in a [^{35}S]GTPγS binding assay[151,152]. Its hydrolysed product, beclomethasone, binds within the orthosteric 7TM pocket of ADGRG3/GPR97 (ref. 152) to induce Go signalling. In mice, beclomethasone dipropionate alleviates psoriasis-like symptoms in WT but not ADGRG3/GPR97-KO animals[66]. Consistent with the inflammatory-regulatory role of ADGRG3/GPR97, the receptor may also mediate a specific effect of curcumin, which activates the receptor in vitro in a cleavage and Stachel-independent manner[239].

Because steroid therapies act broadly via nuclear receptors, using them to modulate aGPCRs carries off-target liabilities, underscoring the need for alternative or improved ligands. Unbiased in vitro screens have identified steroid-mimetic compounds, such as 3-α-acetoxydihydrodeoxygedunin (α-DOG)[240], dihydromunduletone (DHM)[241] and hexahydroquinoline derivatives[242], that modulate signalling via ADGRG1/GPR56 and ADGRG5/GPR114. α-DOG emerged as a partial agonist of ADGRG1/GPR56 in an SRE luciferase screen, activating a disabled ($A^{386}M$) CTF mutant yet inhibiting Stachel peptide-driven responses at higher concentrations[240]. Concentration responses in a G_{13} GTPγS reconstitution assay using the ($A^{386}M$) CTF mutant revealed an EC_{50} of 4.8 μM. The compound also elicited signalling in ADGRG5/GPR114-transfected cells while inhibiting activation of ADGRF1/GPR110. A follow-up study has shown activation of full-length ADGRG1/GPR56 and demonstrated that GPS cleavage is necessary for α-DOG stimulation[171]. Although the α-DOG binding site on ADGRG1/GPR56 is unknown, both studies indicate a requirement for the Stachel sequence, revealing a notable pattern: α-DOG does not activate the full CTF or a completely Stachel-truncated CTF, inhibits an early-Stachel truncation ($Y^{384}M$ CTF), and activates only when the intramolecular Stachel is truncated midway – as in the initially reported $A^{386}M$ CTF and the $F^{385}M$ CTF mutants. These observations suggest that α-DOG requires the lower half of the intramolecular Stachel to bind and activate the full-length receptor and cannot compete with a fully exposed Stachel (as in the CTF following NTF release) or with the synthetic Stachel peptide, and that in cleavage-deficient mutants, the required Stachel region is inaccessible, implying a conformation distinct from undissociated WT full-length receptor, wherein at least part of the Stachel appears accessible. Consistent with this, prior work on ADGRE5/CD97, ADGRL1/LPHN1 and ADGRB3/BAI3 reported that Stachel positions +3 and +6 are solvent-accessible within full-length receptors, potentially through flexible GAIN domain 'flaps'[243]. In the ADGRG1/GPR56 study, the flaps should not differ between WT and cleavage-deficient full-length receptors, pointing instead to access via the GAIN domain–7TMD interface.

DHM was identified as an ADGRG1/GPR56 inhibitor with an IC_{50} of 20.9 μM in the G_{13} GTPγS assay applied to the CTF mutant of the

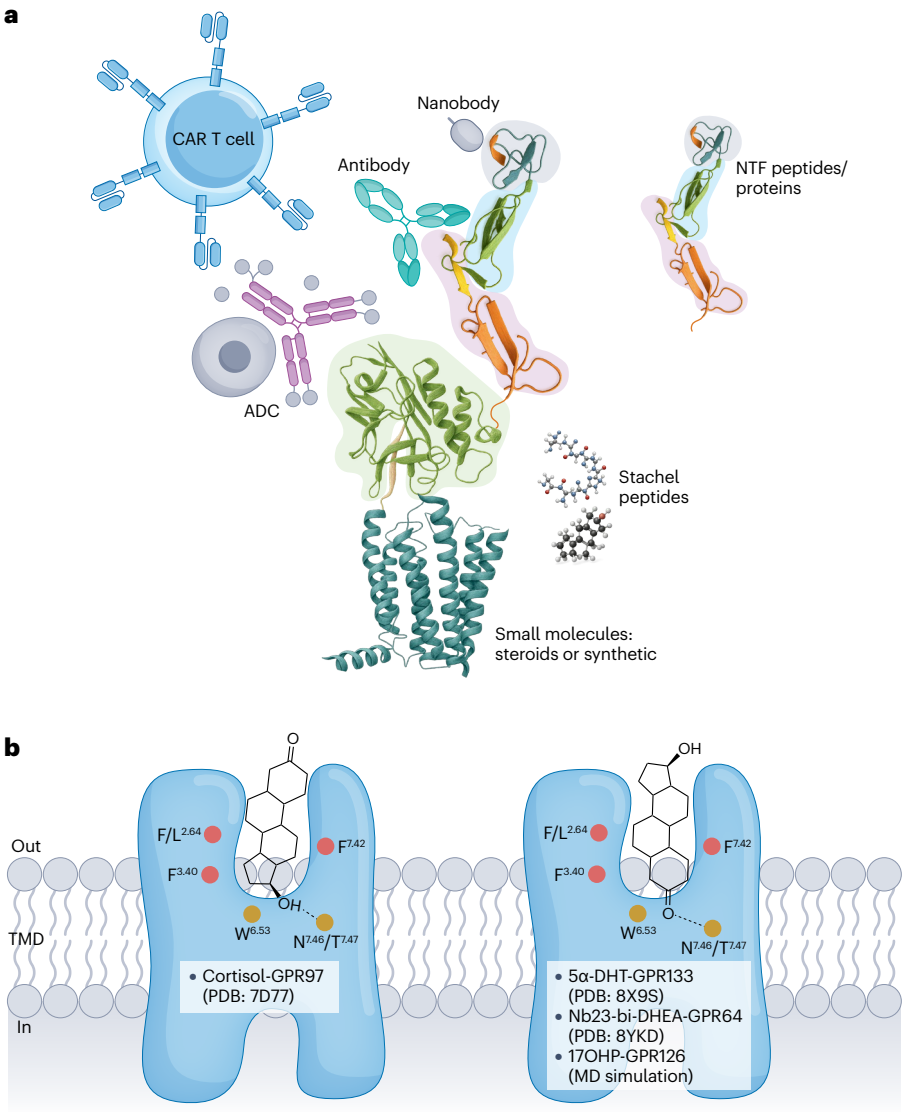


Fig. 4 | Pharmaceutical targeting approaches for adhesion GPCRs. a, The large N terminus of adhesion G-protein-coupled receptors (aGPCRs) presents several promising pharmaceutical approaches, in addition to the classical targeting of the seven-transmembrane domain (7TMD) binding pocket with small molecules or synthetic Stachel peptides. Antibodies and nanobodies are pivotal for detecting the N terminus, which is often prone to splice variation. These can activate the receptor, block interaction with ligands, or be engineered to carry drugs, as seen in antibody–drug conjugates (ADCs), or to guide activated T cells with designed chimeric antigen receptors (CARs). Synthetic peptides that mimic parts of the N-terminal fragment (NTF) can also be used to inhibit processes, such as angiogenesis, as demonstrated by vasculostatins. **b**, Two common contacting motifs, $(F/L)^{2.64}$-$F^{3.40}$-$W^{6.53}$ and $F^{7.42}XXN^{7.46}/T^{7.47}$, within aGPCRs for two distinct orientations of steroid ligands. Key pocket residues are highlighted in red circles, with polar interactions indicated by dashed lines. MD, molecular dynamics.

Glossary

ADGR
Unified gene-symbol system for mammalian adhesion GPCRs (for example, ADGRG1 for GPR56), grouping receptors into nine classes (A–G, L and V) based on sequence and domain architecture.

Adhesion G-protein-coupled receptors
(aGPCRs). A large subfamily of GPCRs characterized by long N termini containing adhesion modules and a GPCR autoproteolysis-inducing (GAIN) domain, involved in development, mechanosensation and disease.

Allosteric modulation
Ligand binding to a site on a receptor that is distinct from the primary (orthosteric) ligand-binding pocket; in aGPCRs, allosteric binding often occurs in the N terminus (for example, extracellular matrix proteins, antibodies and nanobodies) and modulates receptor conformation, Stachel exposure and signalling output without directly occupying the 7TMD orthosteric site.

C-Terminal fragment
(CTF). The membrane-embedded part of an aGPCR generated by GPS/GAIN cleavage; contains the 7TMD, intracellular C terminus, and the N-terminal 'Stachel' sequence; often constitutively active.

GPCR autoproteolysis-inducing domain
(GAIN domain). A conserved extracellular domain in aGPCRs that carries the GPCR proteolytic site (GPS) and mediates autoproteolytic cleavage into N-terminal fragment (NTF) and CTF; also shelters the Stachel sequence.

GPCR proteolytic site
(GPS). Highly conserved cleavage site within the GAIN domain wherein autoproteolysis divides the receptor into NTF and CTF.

N-Terminal fragment
(NTF). Extracellular portion of aGPCRs encompassing the adhesion modules and GAIN domain; remains non-covalently associated with CTF after cleavage; can also be shed and act in *trans*.

Orthosteric pocket
Primary ligand-binding cavity within the 7TMD wherein endogenous ligands (for example, steroids and Stachel) and most small-molecule drugs bind to modulate receptor activity.

Seven-transmembrane domain
(7TMD or 7TM domain). The canonical GPCR core formed by seven-membrane-spanning helices that house the orthosteric ligand-binding pocket and engage G proteins and arrestins.

Stachel sequence
Short intramolecular agonistic sequence at the N terminus of the C-terminal fragment of aGPCRs; it was initially described as a tethered agonist (by analogy to protease-activated receptors), but, to avoid ambiguity with other covalently attached ligands, this Review uses the specific term Stachel for the aGPCR intramolecular ligand and avoids 'tethered agonist'.

***Trans* signalling**
For aGPCRs; signalling events in which aGPCR-derived fragments or complexes modulate other receptors on the same or neighbouring cells, independent of the G-protein coupling of aGPCR.

receptor[241]. It also inhibits full-length ADGRG1/GPR56 and ADGRG5/GPR114 while sparing ADGRF1/GPR110. Although its binding site is unknown, competition with endogenous and synthetic Stachel agonists implies that DHM may act as an orthosteric antagonist of ADGRG1/GPR56. Other isoflavonoids in the same study inhibited ADGRG1/GPR56 CTF activity, but the effects were attributed to cytotoxicity. DHM itself promotes cell death by inhibiting mitochondrial complex I (with an IC_{50} of 146.9 nM), limiting its therapeutic potential. Nevertheless, DHM was tested in a mouse model of hepatocellular carcinoma[244]. WT mice received orthotopic Hep3B cells overexpressing ADGRG1/GPR56 and were treated from weeks 2–6 with galunisertib (GAL; a TGFBR1 inhibitor, 10 μM), DHM (20 μM) or both. Only DHM alone or the combination significantly reduced tumour burden and prolonged survival. However, because neither these in vivo experiments nor in vitro assays controlled for ADGRG1/GPR56 knockdown, receptor-mediated effects cannot be concluded; given the measured 20.9 ± 2.4 μM IC_{50} at the receptor versus the far lower 146.9 nM IC_{50} for mitochondrial inhibition, the latter mechanism is more plausible in vivo at 20 μM DHM. Among hexahydroquinolines, 'compound 4' and 'compound 36' activated ADGRG1/GPR56 Stachel-deficient truncation mutants[242], but only compound 36 achieved sufficient signal saturation to estimate an EC_{50} (2.95 μM), and it activated both cleavage-impaired and Stachel-impaired full-length ADGRG1/GPR56 without inducing signalling in other tested aGPCRs, while significantly inhibiting ADGRG3/GPR97. Virtual docking places compound 36 and its derivative 36.40 in the orthosteric binding pocket of ADGRG1/GPR56.

Finally, in vivo phenotypic screens in hypomorphic *adgrg6/gpr126* zebrafish identified compounds that rescued myelination[25] and ear swelling defects[245,246]. Only the myelination hits were followed by in vitro characterization, wherein most increased cAMP in an ADGRG6/GPR126-independent manner, and only apomorphine hydrochloride showed receptor-mediated activation, with efficacy, but not potency, dependent on the Stachel sequence[25]; its binding mode remains unknown.

The limited receptor specificity of many of these compounds underscores the need for refined, structure-guided approaches that became possible with the advent of several 7TMD structures of aGPCRs featuring the bound Stachel or steroid ligands. Steroids share a hydrophobic tetracyclic core whose stereochemistry and substituents (for example, hydroxyl and keto groups) dictate biological activity. Although the core is hydrophobic, polar moieties increase overall hydrophilicity, and inter-ring motions confer conformational flexibility. These dynamics, governed by dihedral angles at ring junctions, shape the three-dimensional topology crucial for aGPCR recognition. Structural analyses, experimental and computational, show that ADGRG3/GPR97, ADGRG2/GPR64 and ADGRD1/GPR133 each contain a ligand-binding pocket that accommodates their respective steroid ligands[89,152,153] (Fig. 4b). This pocket is formed primarily by transmembrane helices TM1–TM3 and TM5–TM7, together with extracellular loop ECL2. In these receptors, the highly conserved toggle-switch residue $W^{6.53}$ (Wootten numbering) sits at the base of the pocket and forms direct contacts with either the A-ring or D-ring of the steroid. Across receptors, the hydrophobic tetracycle is recognized by a conserved $\Phi(F/L)^{2.64}$–$F^{3.40}$–$W^{6.53}$ motif, wherein $F/L^{2.64}$ and $F^{3.40}$ engage in hydrophobic or π–π interactions with the steroid skeleton. In addition, the aromatic residue $F^{7.42}$ and the polar residue $N^{7.46}$ form an $F^{7.42}XXN^{7.46}$ motif that recognizes hydroxyl and carbonyl substituents, indicating that the aromaticity of $F^{7.42}$ and polarity of $N^{7.46}$ are critical for specific recognition of steroid functional groups.

The selective ADGRD1/GPR133 agonist AP503 illustrates structure-guided discovery[89]. A combined forward and reverse virtual-screening workflow docked approximately 1 million compounds into the ADGRD1/GPR133 pocket, using a predicted polar interaction with $N^{7.46}$ as an initial filter to select the top 1,000 candidates. These were then docked to the androgen receptor (AR), and putative AR binders excluded. After clustering to ensure chemical

diversity and applying an additional filter for interactions with key ADGRD1/GPR133 residues ($Q^{1.36}$, $L^{2.64}$, $F^{2.68}$), 60 candidates remained. Structure–activity relationship optimization yielded AP503, which selectively activated Gs signalling by ADGRD1/GPR133 with an EC_{50} of 1.2 nM, without affecting nuclear ARs or other aGPCR family members, thereby reducing androgen-related adverse effects. A similar approach was used to establish GL64, which displayed an EC_{50} of 4 μM in vitro[91]. AP503 and GL64 showed osteoanabolic activity by activating ADGRD1/GPR133 on osteoblasts and inhibiting octeoclasts, increasing bone mass and alleviating osteoporosis[90,91]. This study offers a template for small-molecule ligand discovery across aGPCRs. Because more than 70% of aGPCRs share these critical residues, the presence of endogenous steroid or steroid-like ligands is plausible, and the conserved motifs provide a structural basis for computer-aided drug design using virtual screening and molecular dynamics simulations.

Prediction of ligands for aGPCRs

Recent structural solutions of aGPCR in complex with steroid hormones and synthetic compounds, along with the structures of their extracellular domains bound to ligands, have laid a crucial foundation for predicting uncharacterized aGPCR ligands. This is especially true when such information is integrated with the application of AlphaFold and other AI-assisted computational methods. For instance, over 70% of aGPCR members share the $F(F/L)^{2.64}$–$F^{3.40}$–$W^{6.53}$ and $F^{7.42}XXN/D^{7.46}$ motifs. These motifs have been shown to have pivotal roles in π–π stacking or hydrophobic interactions with steroid rings, as well as in polar interactions with hydroxyl or carbonyl groups attached to the steroid rings in steroids. Consequently, these aGPCR members may utilize steroids or steroid derivatives as their endogenous ligands. To predict other novel 7TMD ligands for aGPCRs, a PubChem database encompassing natural steroids and a variety of structural derivatives, including oxidized or reduced forms, and esterified and sulfated modifications, could be established. Subsequently, docking screening and molecular dynamics simulations could be used to assess the conformational stability of ligand–receptor complexes and the persistence of key interactions. On the basis of these evaluations, Molecular Mechanics Poisson–Boltzmann Surface Area free energy calculations could be applied to rank candidate compounds according to their binding affinities. This would enable the prioritization of compounds with stronger interactions for subsequent biochemical characterization and functional validation. On the basis of our current knowledge, the prediction and design of novel 7TMD binding agonists for aGPCRs should focus on two aspects: (1) the steroid/Stachel binding pocket harbours hydrophobic residues, which could be utilized to design compounds of a hydrophobic nature with improved ligand affinity, and (2) the less conserved polar residues inside the 7TMD pocket that may be useful for designing interactions that increase the selectivity of the ligands.

In addition to steroids, numerous aGPCRs utilize their extracellular N-terminal regions to recognize matrix proteins or the extracellular domains of other membrane proteins. Presently, with the assistance of structural prediction tools such as AlphaFold, it is feasible to model the conformations of these N-terminal domains. By integrating interface feature analysis and molecular docking, potential protein ligands can be screened at the molecular level. Furthermore, techniques such as molecular dynamics simulations, cross-linking mass spectrometry, and systematic point mutation experiments can be used to predict and validate the interaction interfaces and complex conformations between aGPCR domains and their ligands. Leveraging this foundation, structure-based design strategies present novel opportunities for discovering and optimizing innovative ligands targeting aGPCRs, which mimic the endogenous N-terminal binding partners. For instance, RoseTTAFold diffusion can be utilized to generate initial protein backbone structures under the constraints imposed by the N-terminal architecture of the receptor. This can then be followed by sequence optimization using Protein Message Passing Neural Network to obtain candidate ligands with enhanced stability and high binding affinity. Subsequently, AlphaFold can predict the structures of receptor–ligand complexes, and biochemical experiments can be conducted for validation and refinement, facilitating the stepwise development of reliable complex models. Moreover, on the basis of the resolved aGPCR structures and the structural models predicted by AlphaFold2, potential ligand binding pockets can be pinpointed using software such as CSM-Potential, SiteMap and P2Rank. These identified ligand pockets are suitable for high-throughput virtual screening. Subsequently, candidate compounds are ranked through

Box 1 | Considerations for establishing adhesion GPCRs as drug targets

General (platform-agnostic):

- Solubility determines feasible routes and exposures: optimize early (salt forms, co-solvents, prodrugs or cyclization for peptides) to avoid late-stage formulation risks.
- Toxicity must capture both on-target and off-target liabilities, including immune activation and cardiopulmonary/CNS effects: orthogonal counterscreens should be performed and translational safety biomarkers be determined.
- Pharmacokinetics or pharmacodynamics should account for protein binding, clearance and half-life, as well as target-mediated drug disposition when receptor expression is high or shed domains create an antigen sink: build exposure–response models that reflect tissue distribution. Blood–brain barrier accessibility is pivotal for CNS indications: match modality to delivery strategy (for example, intranasal, intrathecal and receptor-mediated transport) and address efflux liabilities.

Adhesion G-protein-coupled receptor (GPCR)-specific (mechanism-aware):

- Expression and disease-driven shifts of splice variants can alter N-terminal architecture, GPCR proteolytic site (GPS) cleavage, Stachel accessibility and co-receptor interactions, reshaping pharmacology across tissues and species.
- Mechanical forces and endogenous ligands (matrix stiffness and composition, shear, cell–cell adhesion, hormones and lipids) dynamically tune receptor activity, bias and N-terminal fragment–C-terminal fragment (NTF–CTF) equilibria: assays must reproduce these contexts.
- NTF shedding and GPS status influence ligand access and can generate soluble antigen sinks: measure NTF in biofluids to inform dosing.
- Co-factors (for example, tetraspanins CD9/CD81) modulate signalling amplitude and pathway choice: include them in screening. Internalization and trafficking determine durability of effect and enable payload strategies (for example, antibody–drug conjugates or targeted delivery).

scoring functions, pharmacophore analysis, and toxicity predictions. Molecular dynamics simulations are then utilized to assess the stability of ligand–receptor complexes and the durability of key interactions. Meanwhile, free energy calculations (such as MM/GBSA or FEP) offer quantitative evaluations of binding affinity. This comprehensive pipeline lays down a theoretical foundation for refining candidate drugs and steering experimental validation.

Outlook

Although no drugs targeting aGPCRs are currently approved, several aGPCR modulators are available and show varying potential to progress towards clinical use. Although many compounds remain at the initial discovery stage, some have achieved in vivo validation in animal models and, in one instance, evaluation in a human cohort. The future trajectory of aGPCR research holds considerable promise for advancing therapeutic strategies across diverse diseases. As understanding of aGPCR structural and functional intricacies deepens, facilitated by advances in cryo-electron microscopy, there is substantial opportunity to design drugs that are both highly specific and effective, thereby minimizing adverse effects. To realize this potential, general drug development considerations (toxicity, solubility, pharmacodynamics and blood–brain barrier penetration) should be weighed alongside receptor-specific functional and structural features (mechanical force sensitivity, extracellular matrix ligands, endogenous steroid ligands, and splice variation) (Box 1). Continued exploration of the diverse signalling pathways mediated by aGPCRs in physiological and pathological contexts is essential. Furthermore, integrating innovative technologies and leveraging robust animal models will be crucial for translating foundational insights into viable clinical applications.

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